LOCAL GOVERNMENT UNIT 2: LEGISLATIVE AGENDA AND CALENDAR MANAGEMENT SYSTEM WITH AUTOMATED DOCUMENT ANALYSIS AND SUMMARIZATION USING AI-CLASSIFICATION

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ABSTRACT

Title: **LOCAL GOVERNMENT UNIT 2: LEGISLATIVE AGENDA AND CALENDAR MANAGEMENT SYSTEM WITH AUTOMATED DOCUMENT ANALYSIS AND SUMMARIZATION USING AI-CLASSIFICATION**

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The Legislative Agenda & Calendar System with Automated Document Analysis and Summarization Using AI-Classification can be seen as an all-in-one scheduling framework set up for the crucial government process. The system brings together and presides over all important legislative activities such as the plenary sessions, the committee meetings, public hearings, and the consultations of various departments

and groups of the organization. This system, functioning as a focal point, guarantees all participants from the Presiding Officer to the general public, sufficient, correct, and on-time information regarding the timing and location of the coming legislative activities. The main focus of the system is on appropriate timing and good adherence to the deadlines, while at the same time, it is smoothly working with other workflow stages, especially in the areas of ordinance tracking and session management. The legislative agenda and calendar system's combination of centralized scheduling, automated reminders, deadline monitoring, and real-time session tracking results in the transparency of governance, preventing scheduling conflicts and making the legislative process to be smooth and efficient. Different access levels and customized views for administrators, secretariat staff, committee chairs, councilors, and the public are provided by the system which, in turn, promotes communication and coordination within and accountability from the public in the legislative process. The system's multi-layered approach means that different user needs are being addressed while data integrity is being kept and critical deadlines are not missed, thus reinforcing the effectiveness and credibility of the legislative institution.

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Chapter 1

INTRODUCTION

1.1 Background of the Capstone Project

The effective governance of Local Government Units (LGUs) necessitates the proper and timely creation of ordinances and resolutions, which are the main regulatory measures of stakeholders' interests, through efficient legislative processes. However, many of the Philippine legislative offices still use manual and fragmented systems that cause the same problems of delays, inconsistent record-keeping, and legislative measures being tracked with difficulty. Furthermore, the amount of legislative documents often exceeds the capacity of the administrative staff who are skilled in various functions but not in legal document handling thus making it difficult for them to carry out analysis and extraction of key information. These limitations lead to a lack of governance effectiveness and difficulty in determining whether local laws are actually achieving the desired results or not.

The City Council of Valenzuela is going through these situations mainly concerning agenda scheduling, legislative tracking, and document management. In an interview with **Mr. Jojo**

Faustino, the Administrative Staff of the Sangguniang Panlungsod Office, some problems that were mentioned are: there is no unified digital system, managing the workload of committees is challenging, meeting and session scheduling is inefficient, and the manually processing of legislative documents is a delay. These problems indicate the need for an organized and tech-enabled approach to managing legislative operations that is quite urgent.

The Local Government Unit Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification is the solution to these issues. This integrated system will be the main scheduling mechanism for legislative workflow, handling plenary sessions, committee meetings, public hearings, and stakeholder consultations, while simultaneously analyzing and classifying the legislative documents. By simplifying the legislative processes and document management together on one digital platform, the system ensures that schedules are followed, deadlines are met, and critical information from documents is automatically extracted and summarized.

1.2 Context and Scope

The local government institutions at the municipal level have a crucial influence on making decisions, enacting laws and passing resolutions which build community directly. With respect to the Valenzuela City Sangguniang Panlungsod, the legislative workflow has a number of closely related activities like planning the agenda, deciding when sessions will be held, coordinating committees, and managing the paperworks. Most of these activities are still done either manually or through separate systems that results in a slow process that is prone to mistakes and takes a lot of time to solve issues.

This project is proposing in response to these challenges, the development of a Local Government Unit 2: Legislative Agenda and Calendar Management System with Automated Document Analysis Using AI-Classification. This system will consist of several modules that can communicate with one another and are capable of managing the calendar, scheduling sessions, creating agendas, monitoring deadlines, and sending automated reminders.

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The scope of this project focuses on the design and development of an integrated legislative agenda and calendar management system intended for use by the Valenzuela City Sangguniang Panlungsod. The system will serve as a centralized platform for managing all legislative schedules, ensuring that calendar information remains consistent and accessible to all authorized stakeholders.

1.3 Problem Statement

The legislative operations of the Valenzuela City Sangguniang Panlungsod involve numerous processes such as agenda scheduling, meeting coordination, ordinance tracking, and document management. However, despite the importance of these functions, the current workflow remains heavily dependent on manual procedures and traditional paper-based systems. This results in several operational challenges that we found. During the interview with Mr. Jojo Faustino, Administrative Staff under the Sangguniang Panlungsod Secretary, key issues were identified that highlight the need for a modernized and centralized digital solution.

Problem	Description
Agenda Scheduling	Manual and inconsistent agenda scheduling processes create scheduling

	conflicts, missed deadlines, and unclear communication about legislative event timelines, making it difficult to coordinate activities effectively.
Activity Coordination	Difficulty in coordinating multiple levels of legislative activities including plenary sessions, committee meetings, public hearings, and consultations results in inefficient workflow and missed opportunities for stakeholder engagement.
Notification Systems	Fragmented notification systems fail to reliably alert stakeholders, committee members, and legislative staff about upcoming meetings, deadlines, and required actions in a timely manner, leaving people unprepared and deadlines slipping.
Information Access	Lack of a centralized platform where legislative staff, committee members, and stakeholders can access updated and accurate information regarding scheduled events, deadlines, ordinance status, and legislative progress, forcing people to ask multiple sources for information and creating inconsistency.
Deadline Management	Absence of automated deadline tracking and conflict prevention tools leads to missed legal requirements, delayed readings, and publication date conflicts that could invalidate legislative work or create legal problems.
Document Processing	Manual document processing and analysis requires staff to spend excessive time reviewing, categorizing, and summarizing legislative documents without systematic support or

	organization, consuming resources that could be spent on substantive legislative work.

Table no.1 Problem Statement

1.4 Objectives and Goals

The primary objective of this capstone project is to design and develop a centralized LGU Legislative Agenda and Calendar Management System with Automated Document Analysis Using AI-Classification, capabilities that enhances the efficiency, coordination, and organization of legislative scheduling and document management within the Valenzuela City Sangguniang Panlungsod.

1. To develop and implement a centralized scheduling platform that serves as a single source of truth for all legislative events including plenary sessions, committee meetings, public hearings, and consultations. This goal will eliminate scheduling conflicts and ensure clear communication across the legislative body.

2. Create an automated deadline tracking and management functionality that systematically monitors legal deadlines, reading

schedules, publication dates, and other critical timeline requirements for ordinances and resolutions. This goal will prevent missed deadlines and procedural errors that could invalidate legislative work.

3. To design and integrate legislative workflow stages—including First Reading, Second Reading, and Third Reading—with the calendar and agenda system to provide seamless coordination between scheduling and ordinance progression. This goal will ensure proper sequencing and timing of legislative processes from proposal through final implementation.

4. To develop an automated notification and reminder system that alerts stakeholders, committee members, and legislative staff about upcoming events, approaching deadlines, and required actions in a timely manner. This goal will improve responsiveness and accountability throughout the organization.

5. To build automated document analysis and summarization capabilities using AI-classification technology that categorizes, processes, and summarizes legislative documents such as bills, resolutions, committee reports, and meeting minutes. This goal will

reduce manual review time and improve information accessibility for all users.

6. To implement conflict prevention mechanisms that automatically detect and alert staff to scheduling conflicts, overlapping meetings, or competing deadlines across different legislative activities. This goal will prevent operational disruptions and ensure smooth workflow execution.

1.5 Significance and Relevance

The successful development and implementation of the Local Government Unit Legislative Agenda and Calendar Management System with Automated Document Analysis Using AI-Classification are highly significant as a response to the digital transformation mandate within the Philippine government. This project is directly relevant as it provides a practical, automated solution to real-world challenges in legislative scheduling and document management.

Valenzuela City Sangguniang Panlungsod: Gaining streamlined legislative operations with centralized calendar management, automated document processing that accelerates ordinance and resolution tracking, and immediate access to organized legislative schedules. This strengthens the legislative body's capacity to

respond to community needs while ensuring compliance with legal deadlines and procedural requirements.

LGU Administrative Staff: Minimizing manual, repetitive tasks such as calendar maintenance, schedule tracking, and notification management, allowing personnel to focus on strategic support and higher-value legislative activities. This increases productivity, reduces errors in scheduling and deadline management, and improves job satisfaction by eliminating time-consuming administrative burdens.

Valenzuela City Constituents: Benefiting from a more responsive and accountable local government through improved efficiency in processing ordinances that address community needs. The public transparency feature enables citizens to access legislative schedules and monitor ordinance progress, fostering greater civic engagement and trust in local governance.

Other Philippine Local Government Units: Using this system as a replicable and tested model for modernizing legislative operations. The comprehensive documentation and architecture provide a practical template for digitalizing legislative departments,

contributing to advancing digital governance standards and accelerating the nation's e-governance agenda.

Future Researchers and IT Professionals: Utilizing this project as a foundational resource demonstrating the complete system development lifecycle, providing empirical evidence on efficiency gains of legislative workflow automation, and informing future research in IT security, authentication protocols, and technology adoption models in government settings.

1.6 Structure of the Document

This capstone project is organized into three chapters, following the standard format for an academic thesis in Information Technology. Each chapter serves a specific purpose in presenting the problem, the solution (the developed system), the methodology, the results, and the conclusions.

Chapter 1: Introduction

This initial chapter lays the foundation for the entire study by introducing the problem and defining the boundaries of the research. It consists of the following sections:

1.1 Background of the Capstone Project

Provides the global, national, and local context of the study, detailing the shift toward e-governance and digital transformation across the Philippine government. This section pinpoints the operational challenges within the Valenzuela City Sangguniang Panlungsod, specifically related to legislative scheduling, deadline management, and document processing. It establishes the urgency of implementing a modernized, centralized system to support legislative operations.

1.2 Context and Scope

This section defines the study boundaries within the Valenzuela City Sangguniang Panlungsod involving administrative staff, committee members, and constituents. The system includes centralized scheduling, workflow integration with ordinance reading stages, deadline management, multi-level coordination, automated reminders, conflict prevention, and public transparency features. Key exclusions are financial management, performance analytics, and constituent engagement portals, keeping focus on core scheduling and document management functions.

1.3 Problem Statement

This section formalizes the research gap by detailing the specific issues identified in the current manual and fragmented legislative processes of the Valenzuela City Sangguniang Panlungsod.

1.4 Objectives and Goals

Presents the overall General Objective of the study: to develop a centralized and automated legislative scheduling and calendar management system that coordinates all legislative events, manages deadlines, automates document processing, and enables systematic tracking of ordinances. This section also outlines ten specific, measurable objectives that the system must achieve, including establishing centralized scheduling, automating deadline tracking, integrating legislative workflow stages, implementing automated notifications, developing document analysis capabilities, creating conflict prevention mechanisms, ensuring public transparency, implementing access control, maintaining a digital repository, and designing a user-friendly interface.

1.5 Significance and Relevance

This section articulates the study's contribution to academic knowledge and its practical benefits for key beneficiaries. It details how the system benefits the Valenzuela City Sangguniang Panlungsod by streamlining operations and enhancing transparency, how administrative staff gain improved productivity and reduced workload, how constituents benefit from more responsive and accountable government, how other Philippine LGUs can replicate the system as a tested model, and how future researchers gain foundational resources for advancing E-Governance and Information Technology research.

Chapter 2: Review of Related Literature - This chapter analyzes existing literature on legislative agenda and calendar management.

Chapter 3: Methodology – The study underlines the research methodology, indicating how the Agile Scrum framework will be used to ensure that the development is structured and a collaboration between teams takes place. It outlines the team roles, sprint cycles, and the use of system design principles, including

microservices and DevOps, in building a flexible, scalable, and efficient records management system.

Chapter 2

REVIEW OF RELATED LITERATURE

2.1 Agile Scrum Methodology Overview

The development of the Local Government Unit Legislative Agenda and Calendar Management System with Automated Document Analysis Using AI-Classification requires a flexible yet structured approach capable of adapting to the specialized requirements of the Valenzuela City Sangguniang Panlungsod. For this reason, the Agile Scrum Methodology was selected as the primary framework, moving away from rigid, sequential development models. Scrum is a lightweight, iterative, and incremental framework designed to manage complex product development efficiently, prioritizing working software and collaborative engagement over extensive planning and formal processes.

Scrum is grounded in the Agile Manifesto, which establishes four core values that are highly applicable to developing solutions for public sector organizations. The first value emphasizes individuals and interactions over processes and tools, meaning that direct communication between developers and stakeholders takes priority over rigid procedures and tools. The second value prioritizes

working software over comprehensive documentation, ensuring that functional system components that users can test and evaluate are prioritized over extensive written documentation. The third value highlights customer collaboration over contract negotiation, where active partnership with stakeholders drives development rather than fixed contractual agreements. The fourth and final value values responding to change over following a plan, recognizing that the flexibility to adapt to new insights and changing requirements is valued over strict adherence to predetermined plans. Together, these four values create a framework that enables development teams to build systems that truly meet user needs and respond to real-world challenges.

The principle of "Responding to change" is particularly important in this project, as requirements gathered through interviews with legislative staff such as Mr. Jojo Faustino are often refined and clarified as they interact with working prototypes. The iterative and incremental nature of Scrum, where the Legislative Agenda and Calendar Management System is developed in short cycles called Sprints, ensures that user feedback is continuously integrated into the system design. This approach minimizes the risk

of delivering a system misaligned with the actual operational needs and workflows of the Sangguniang Panlungsod.

2.1.2 The Scrum Framework: Roles, Artifacts, and Events

The Scrum framework provides a structured approach for managing the development of the Legislative Agenda and Calendar Management System through clearly defined roles, artifacts, and events. The Development Team is self-organizing and cross-functional, responsible for designing, developing, and delivering working increments of the system. The team possesses the technical skills necessary to build scheduling modules, implement deadline tracking features, and develop document analysis capabilities. The Product Owner represents the voice of the customer, in this case the Valenzuela City Sangguniang Panlungsod Secretary's Office, and is responsible for defining system features, prioritizing backlog items such as centralized scheduling, automated reminders, conflict prevention, and document summarization capabilities, and ensuring that development efforts align with the organizational needs of the legislative body. The Scrum Master serves as a facilitator and coach, ensuring that the development team adheres to Scrum principles and practices by removing impediments that may delay

development, facilitating communication between team members and stakeholders, and maintaining focus on sprint goals and timelines.

The framework also establishes key artifacts that guide development. The Product Backlog is a comprehensive, prioritized list of all features, functions, and requirements for the system, including user authentication and role management, centralized calendar interface, automated deadline tracking, notification system, conflict detection algorithms, document analysis module, and public transparency features. The Sprint Backlog represents the subset of Product Backlog items that the Development Team selects and commits to completing during a single, time-boxed Sprint typically lasting two to four weeks. The Increment is the working, potentially releasable software delivered at the end of each Sprint, which could be a fully functional module such as the centralized calendar interface, the automated deadline tracking system, or the document analysis and summarization feature, all of which can be demonstrated to stakeholders and used for gathering feedback.

The development cycle is structured by key events designed for team synchronization, stakeholder engagement, and continuous improvement. Sprint Planning is held at the beginning of each Sprint

to define the Sprint Goal and select backlog items to be completed, with sprint goals such as "Implement and test the centralized scheduling module with basic conflict detection." The Daily Scrum is a brief daily meeting, typically 15 minutes, where team members synchronize activities, discuss progress, and identify any obstacles that need to be addressed within the next 24 hours. The Sprint Review is held at the end of each Sprint, where the Development Team demonstrates the completed Increment to the Sangguniang Panlungsod staff and other stakeholders, which is critical for gathering feedback on usability, functionality, and alignment with legislative workflows to inform priorities for subsequent sprints. The Sprint Retrospective is an internal team meeting held after the Sprint Review, where the Development Team reflects on the past Sprint, discussing what went well, what could be improved, and what process changes will be implemented in the next Sprint, ensuring that the team becomes more efficient and productive over time through continuous improvement.

2.1.3 Application to Legislative Agenda and Calendar Management System Development

The selection of Scrum is strategically aligned with the complexity of managing legislative scheduling, deadline tracking,

and document processing. By developing the system in stages through iterative sprints, the team can ensure that critical features such as automated deadline tracking, conflict prevention, and document analysis are refined based on practical use and stakeholder feedback. This approach guarantees that the final system is not only technically functional but also highly usable and accurately reflects the complex procedural requirements, operational workflows, and specialized needs of the Sangguniang Panlungsod. The continuous feedback loop between developers and legislative staff throughout the project lifecycle results in a system that truly serves the organization's mission of efficient and transparent governance.

2.2 Enterprise Architecture Concepts

Building the Legislative Agenda and Calendar Management System with Automated Document Analysis Using AI-Classification is more than just writing code. We're essentially redesigning how an organization manages its core processes. That's where Enterprise Architecture comes in. It provides a framework for thinking about how IT systems fit into the bigger organizational picture, how technology actually supports what the government is trying to accomplish. Enterprise Architecture is a blueprint for how an

organization operates. In theory, anyway. In practice, it's about making sure that when you build an IT system, it actually aligns with the organization's real goals and needs. For the Sangguniang Panlungsod, that means the Legislative Agenda and Calendar Management System needs to fit into how the city government operates.

The field typically breaks down into four domains. Business Architecture is about understanding the actual legislative process—how ordinances move through the system, which committees review what, understanding the workflow, the procedures, the real operations. For the Sangguniang Panlungsod, it's understanding how Mr. Jojo Faustino and his team actually work, what they need to accomplish each day. Data Architecture focuses on where information lives and what data needs to be stored and managed. Currently, legislative information is scattered everywhere across paper files, email, and personal notebooks. The centralized database for the system will store schedules, ordinance records, deadlines, and documents all in one place, which is huge compared to the current situation. Application Architecture is about how the different components talk to each other. The scheduling module needs to work with deadline tracking. Document analysis needs to

feed information to the calendar. It's about making sure pieces don't just exist independently but actually work as a system. Technology Architecture is the foundation layer covering what databases, programming languages, servers, and infrastructure are needed. It's less interesting conceptually but absolutely critical to actually getting things built.

A big principle in Enterprise Architecture is consolidating fragmented systems into integrated platforms, which is exactly what we're doing here though the Sangguniang Panlungsod's situation is actually pretty extreme since they're coming from mostly manual, paper-based processes. Massive organizations use comprehensive frameworks like TOGAF for managing hundreds of interconnected systems. That's overkill here. We're working with something much more focused. Integration means all legislative data—schedules, ordinances, committee reports, deadlines, documents—lives in one authoritative database. Right now it's everywhere and nowhere. Centralization eliminates the data duplication problem where the same information gets recorded differently in different places, reduces errors, and means when someone updates information, everyone sees the same thing. Interoperability is really just systems talking to each other. The scheduling module shares data with

deadline tracking. The document analysis tool provides summaries that appear in the calendar. When committees update their meeting schedule, the deadline system automatically adjusts. Without this integration, you still end up with disconnected applications that require staff to manually move information between systems, which defeats the purpose of having a centralized system in the first place.

2.3 Microservices Architecture

The Legislative Agenda and Calendar Management System is built on a microservices architecture rather than a traditional monolithic design. This approach breaks the system into smaller, independent services that each handle specific functions. It's a deliberate choice made for scalability, flexibility, and efficiency.

2.4 DevOps and CI/CD

Building the Legislative Agenda and Calendar Management System is one thing. Keeping it running, fixing problems, and making improvements without disrupting the Sangguniang Panlungsod's work is another. That's where DevOps and Continuous Integration/Continuous Deployment (CI/CD) come in. These practices make software development faster, more reliable, and safer. For the Legislative Agenda and Calendar Management

System at Valenzuela City, they enable the team to develop efficiently, maintain the system smoothly, and respond quickly when issues pop up.

DevOps breaks down the wall between developers who build software and operations people who maintain it. Traditionally, these teams worked separately, developers would hand off code to operations, who'd try to deploy it, discover problems, and hand it back. It was slow and frustrating for everyone. DevOps says: work together from the start. The approach focuses on three main things. Automation means stop doing repetitive tasks manually. When a developer makes a change, the system automatically builds it, tests it, and reports results. For the Legislative Agenda and Calendar Management System, this means if someone updates the scheduling module, automated tests immediately verify it doesn't break deadline tracking or document retrieval. Automation catches problems fast and removes human error from routine processes. Rapid Feedback means when something goes wrong, find out immediately. Modern monitoring watches the system 24/7, detecting errors, performance problems, or security issues the moment they occur. The development and operations teams get alerts right away. For the Sangguniang Panlungsod, this means if the calendar

interface becomes slow or the notification system starts failing, someone knows about it within minutes, not hours or days. Quick detection means quick fixes. Continuous Improvement means software delivery isn't static. The team regularly reviews what's working, what isn't, and what could be better. These insights drive improvements to the development process itself. Maybe automated tests could catch more issues. Maybe deployment procedures could be faster. Maybe team communication needs adjustment. DevOps culture encourages constantly asking "how can we do this better?"

Continuous Integration means developers don't work in isolation. Instead, they frequently merge their code changes into a shared system—sometimes multiple times per day. Every time code is merged, automated tests run immediately to catch integration problems. Here's why this matters: imagine one developer is updating the deadline tracking feature while another is improving the notification system. Without CI, they might work separately for weeks, then try to combine their changes and discover they conflict or break each other. That's painful to fix and causes delays. With CI, they merge code frequently. If there's a conflict or problem, it shows up immediately when there's less code to sort through. For the Legislative Agenda and Calendar Management System, CI means

the scheduling module, document analysis service, and calendar interface are constantly tested together. When someone updates how the system handles ordinance deadlines, automated tests verify that scheduling still works, notifications still fire at the right times, and document analysis still produces accurate summaries. If something breaks, the developer knows immediately and can fix it while the problem is fresh in their mind.

2.5 Relevant Studies and Research

This section reviews local and foreign literature that substantiates the need for digital transformation in legislative bodies.

Local Studies

Pernia, R. A. (2023), Pernia (2023) conducted a Philippine-based study using automated text analysis techniques, including keyword extraction and computational text classification, to examine patterns and themes in Philippine political research publications. The study applied Natural Language Processing (NLP) methods to analyze large volumes of political texts, allowing systematic identification of dominant topics and trends without manual reading.

University of the Philippines Center for Integrative and Development Studies (UP-CIDS) (2023), UP-CIDS (2023)

documented discussions from a Philippine academic webinar that explored the use of text mining, machine learning, and automated text analysis in political science and public policymaking. The discussions highlighted real-world applications of AI tools for analyzing government documents, policy texts, and institutional records to support governance and administrative efficiency.

University of the Philippines Center for Integrative and Development Studies (UP-CIDS) (2024), The UP-CIDS (2024) concept note discusses the use of text mining and automated document analysis in policymaking and governance research in the Philippine context. It explains how AI-driven techniques such as text classification, clustering, and summarization can be used to process large sets of policy and government documents efficiently. The paper emphasizes the role of computational tools in improving evidence-based decision-making.

Foreign Studies

Bryłkowski & Klikowski (2025), This study uses large language models (LLMs) to analyze legislative texts from the Polish Parliament, conducting automated content analysis on bills and legal proposals. Although not explicitly about calendar

management, it shows how AI classification and NLP processing can systematically analyze legislative documents, which is essential for building tools that can help with agenda profiling, topic tagging, and tracking legislative priorities over time. By processing legislative corpora with AI, researchers can identify themes relevant to scheduling and prioritization within a legislative calendar.

Gesnouin et al. (2024), This work demonstrates a state-of-the-art AI model designed specifically to generate concise summaries of legislative proposals. Summarization is a key component of automated document analysis, enabling lawmakers and analysts to digest lengthy legislative texts quickly. In the context of legislative agenda and calendar management, such summarization can help identify priority items, track progress, and align documents with scheduled session activities.

Pressato et al. (2025), This paper presents an automated approach using NLP to organize and analyze amendments in legislative documents. It uses topic clustering and retrieval, which directly aligns with classification of agenda items and legislative texts. This kind of processing supports automated summarization and categorization — foundational for tools that would assist

legislative agenda formation and tracking through the calendar of sessions.

International Parliamentary Union (2024), This use case demonstrates how AI-powered summarization is applied to legislative bills and parliamentary documents to make them easier to comprehend. While not a formal academic study, it is an institutional example of AI summarization in a legislative context, bridging automated analysis with legislative workflows like agenda publication and session planning.

OECD / UNESCO Report – Digital Government (2024), Although broader in scope, this OECD report shows how AI can be used to process and categorize large volumes of government text such as statutes, reports, and legislative records. It explicitly discusses categorization and summarization of documents, which can be extended to the legislative domain, supporting automated analysis for legislative agendas and scheduling.

2.6 Integration of Information Systems in Enterprise Environment

In a complex organization like the Valenzuela City government, the effective management of information requires

systems to work together rather than existing in isolated silos. Integration of Information Systems (IIS) refers to the process of connecting disparate software applications and subsystems, both physically and functionally, to allow them to communicate and share data seamlessly. The Legislative Agenda and Calendar Management System is designed not as a passive scheduling tool, but as an active integration hub that connects the entire legislative workflow from planning through public accountability. By integrating with ordinance tracking, committee management, hearing management, and session operations, the system creates a unified legislative lifecycle where data flows continuously from initial agenda planning through final public reporting.

The principles of Enterprise Integration are directly applied to the Legislative Agenda and Calendar Management System to create this unified lifecycle. The system operates across three distinct phases of legislative operations. The Planning Phase involves the Record Department and administrative staff who prepare ordinances and materials for legislative consideration. The system integrates with Group 4's ordinance tracking system, allowing administrators to import ordinance data and reading stage information directly into the Session Scheduling module. This

seamless input integration reduces manual data entry errors, ensures that the ordinances scheduled for debate match the official records stored in the central repository, and eliminates duplicate encoding across multiple offices. When an administrator selects ordinances for a legislative session, the system automatically pulls the correct versions and stages, ensuring consistency between what is debated in the session hall and what is officially recorded.

The Coordination Phase represents the actual legislative process where sessions, committee meetings, and hearings occur. The system achieves Horizontal Integration by bridging the gap between physical calendar management and operational workflows. When the Session Scheduling module creates a legislative session, the Calendar Management module instantly updates to display the event, the Deadline Tracking module automatically calculates compliance deadlines based on the reading stage, and the Reminder System schedules notifications for all stakeholders. This integration removes the need for separate manual scheduling processes or duplicate calendar entries, ensuring that committee chairs, councilors, and administrative staff all see the same information and deadlines. The Agenda Creation and Management module pulls ordinances and committee reports from integrated

systems, attaches supporting documents from the Record Department, and publishes the complete agenda simultaneously through multiple channels. This coordinated approach ensures the Coordination Phase is automated and synchronized, with all stakeholders receiving consistent information about what will be discussed, when, and with what supporting materials.

Chapter 3

METHODOLOGY

3.1 Agile Scrum Methodology in the Project

For our Legislative Agenda and Calendar Management System, we utilize the Agile Scrum methodology; it is a flexible and team-driven approach used to develop a complex legislative information system. This project promotes adaptability, collaboration, and continuous delivery, which are essential in building an AI-enhanced system within a legally structured and procedurally dynamic government environment. The Scrum framework fosters collaboration among cross-functional teams, enabling continuous stakeholder feedback and frequent system iterations

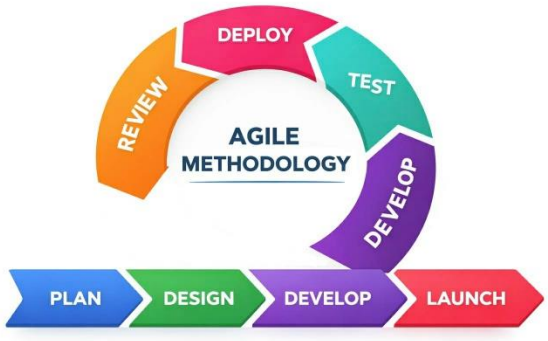


Figure no. 1 Agile Scrum Framework

3.2 Roles and Responsibilities

Each role in the project team carries significant responsibilities that directly influence the success and overall performance of the capstone project. The table below outlines the key duties assigned to each team member:

Name	Role	Responsibilities
Mr. Ronald Roldan Jr.	Research Adviser	Research Adviser for the capstone project. To provide academic guidance and technical oversight to ensure the system meets the required standards of the curriculum, with critiquing the research methodology, reviewing the documentation for accuracy and clarity, and ensuring that the project objectives align with the institutional requirements.
Mr. Jojo Faustino	Product Owner	Directly representing the stakeholders of Valenzuela City Hall. Defining the specific requirements of the *Legislative Agenda and Calendar Management System*

Mr. John Marvin I. Agno	Project Manager	Overseeing the entire software development lifecycle. He is responsible for Sprint Planning, Daily Stand-ups, and Sprint Reviews, to keep the team aligned. Also, manages the project's resources and timeline, working to identify and remove any technical impediments that could hinder the development team's progress.
Mr. Shane Z. Salido	Developer/Lead Programmer	Serves as the Lead Programmer, responsible for the core architectural design and backend development of the system. Manages the database structure, API integrations, and critical security features. Additionally, supervises the coding standards of the development team to ensure the system is scalable and performance optimized.
Mr. Jaymar S. Rabino	Programmer	Focusing on assisting the lead programmer with debugging and testing specific modules. Also supporting documentation specialists to finalize all documents.
Ms. Jamille V, Francisco	Documentation Specialist	Working with other documentation specialists to support the research and writing process. She's responsible for gathering data for building the different types of diagrams and tables.
Mr. Reinhard D. Ganal	Documentation Specialist	Tasked with compiling and organizing all written deliverables for the capstone project. Ensures that the manuscript follows the required academic formatting and citation standards (APA).

Table no.2 Roles and Responsibilities

3.3 Sprint Cycles

The development of the Legislative Agenda and Calendar Management System followed an Agile Scrum methodology, utilizing **Sprint Cycles** to break down complex requirements into manageable tasks. The Scrum Board below visualizes the workflow status during the active development phase. It categorizes features into **"To Do"**, **"In Progress"**, and **"Done"**. This visual tracking ensured that the team maintained a steady velocity and addressed critical dependencies.

To Do	In Progress	Done
• Committee Management Integration • Granular Access Permissions • Agenda Version Control	• Calendar Export Functionality • Comprehensive Audit Logging	• User Authentication System • Basic Calendar Display • Session Timeout & Login Logging • Session Creation & Management

		• Data Encryption Implementation • Responsive Navigation & Layout • Dashboard with Key Metrics • Automated Backup System

Table no. 3 Scrum Board

3.4 Scrum Artifacts

3.4.1 Product Backlog (User Stories)

User Story No.	Features/ Task	User Stories	Priority	Status
MODULE 1: Calendar Management				
US-CAL 01	**Create Calendar**	As an admin, I want to create a legislative calendar so that I can organize all sessions for the year	**High**	

US-CAL 02	**Edit Calendar Entry**	As an admin, I want to edit calendar entries so that I can update session dates when needed	**High**	
US-CAL 03	**Delete Calendar Entry**	As an admin, I want to delete calendar entries so that I can remove cancelled session	**Medium**	
US-CAL 04	**View Calendar**	As a staff, I want to view the legislative calendar so that I can see upcoming sessions	**High**	
US-CAL 05	**View Public Calendar**	As a user, I want to view public calendar so I can track when sessions occur	**Low**	
US-CAL 06	**Set Recurring Sessions**	As an admin, I want to set recurring	**Medium**	

		sessions schedules so that regular meetings are automatically added to the calendar		
US-CAL 07	**Color-code Events**	As an admin, I want to color-code different session types so that I can quickly identify regular, special, and emergency sessions	**Medium**	
US-CAL 08	**Export Calendar**	As a staff, I want to export the calendar to PDF so that I can print it for reference	**Low**	
US-CAL 09	**Switch Calendar Views**	As an admin, I want to switch between monthly, weekly, and daily calendar views so that I can manage	**High**	

		schedules at different granularities		
US-CAL 10	**Filter Calendar**	As a staff, I want to filter calendar events by session type and status so that I can focus on specific activities	**Medium**	
US-CAL 11	**Link Calendar to Sessions**	As an admin, I want calendar entries to automatically link to sessions details so that I can quickly access full information	**High**	
US-CAL 12	**View Calendar Conflicts**	As an admin, I want to see calendar conflicts when scheduling so that I can avoid double-booking dates	**Medium**	

MODULE 2: Session Scheduling				
US-SES-01	**Schedule New Session**	As an admin, I want to schedule new legislative sessions so that I can set the date, time, venue, and type	**High**	
US-SES-02	**Reschedule Session**	As an admin, I want to reschedule sessions so that I can adjust for conflicts or changes in availability	**High**	
US-SES-03	**Cancel Session**	As an admin, I want to cancel sessions so that stakeholders are notified, and calendar is updated	**Medium**	
US-SES-04	**Assign Session Type**	As an admin, I want to assign session types (regular,	**High**	

		special, emergency) so that sessions are properly categorized		
US-SES-05	**View Session Details**	As a staff, I want to view session details so that I can prepare materials and know what to expect	**High**	
US-SES-06	**Set Attendance Requirements**	As an admin, I want to set session attendance requirements so that quorum is tracked and enforced	**Medium**	
US-SES-07	**Add Session Notes**	As an admin, I want to add session notes and remarks so that important details are recorded for reference	**Medium**	

US-SES-08	**Receive Session Confirmation**	As a staff, I want to receive session confirmations so that I know when and where to attend	**High**	
US-SES-09	**View Published Sessions**	As a user, I want to view published session schedules so that I can attend public hearings	**Low**	
US-SES-10	**Attach Documents to Session**	As an admin, I want to attach documents to sessions so that materials are available to attendees	**High**	
US-SES-11	**Track Session Attendance**	As an admin, I want to track session attendance so that I can generate attendance reports	**Medium**	

US-SES-12	**Mark Session Complete**	As an admin, I want to mark sessions as complete so that the system tracks which sessions have concluded	**High**	
US-SES-13	**View Session History**	As a staff, I want to view session history so that I can reference past meetings and decisions	**Medium**	
MODULE 3: Agenda Creation and Management				
US-AGN-01	**Create New Agenda**	As an admin, I want to create a new agenda so that I can organize items for an upcoming session	**High**	
US-AGN-02	**Add Agenda Items**	As an admin, I want to add agenda items so that I can include	**High**	

		ordinances, resolutions, and reports to be discussed		
US-AGN-03	**Reorder Agenda Items**	As an admin, I want to reorder agenda items using drag-and-drop so that priority items are listed first	**High**	
US-AGN-04	**Edit Agenda Items**	As an admin, I want to edit agenda items so that I can update descriptions, details, and assignments	**High**	
US-AGN-05	**Delete Agenda Items**	As an admin, I want to delete agenda items so that I can remove irrelevant or duplicate items	**Medium**	

US-AGN-06	**Attach Documents to Items**	As an admin, I want to attach documents to agenda items so that supporting files are accessible via hyperlinks	**High**	
US-AGN-07	**Set Item Status**	As an admin, I want to set agenda item status (pending, approved, deferred, rejected) so that progress is tracked	**High**	
US-AGN-08	**Assign to Committee**	As an admin, I want to assign agenda items to committee members so that responsibilities are clear	**High**	
US-AGN-09	**View Assigned Items**	As a staff, I want to view my assigned agenda items so that	**High**	

		I can prepare my committee work		
US-AGN-10	**Publish Agenda**	As an admin, I want to publish agendas so that they are visible to staff and the public	**High**	
US-AGN-11	**Add Comments to Items**	As a staff, I want to add comments to assigned agenda items so that I can provide input and feedback	**Medium**	
US-AGN-12	**View Published Agendas**	As a user, I want to view published agendas so that I can see what will be discussed in sessions	**Low**	

US-AGN-13	**Duplicate Previous Agenda**	As an admin, I want to duplicate previous agendas so that I can reuse templates for similar similar sessions	**Medium**	
US-AGN-14	**Archive Old Agendas**	As an admin, I want to archive old agendas so that the system remains organized and clutter-free	**Medium**	
US-AGN-15	**Print Agenda**	As an admin, I want to print agendas in as formatted layout so that I can distribute hard copies during sessions	**Low**	
MODULE 4: Deadline Tracking and Monitoring				
US-DED-01	**Set Deadlines**	As an admin, I want to set deadlines for	**High**	

		agenda items and tasks so that timelines are enforced		
US-DED-02	**View All Deadlines**	As an admin, I want to view all upcoming deadlines in one dashboard so that I can monitor overall progress	**High**	
US-DED-03	**View Overdue Items**	As an admin, I want to see overdue items highlighted in red so that I can follow up on delays immediately	**High**	
US-DED-04	**View My Deadlines**	As a staff, I want to view my assigned deadlines so that I can manage my workload effectively	**High**	

US-DED-05	**Extend Deadlines**	As an admin, I want to extend deadlines so that I can accommodate legitimate delays or changes	**Medium**	
US-DED-06	**Mark as Completed**	As a staff, I want to mark my deadlines as completed so that status is updated in the system	**High**	
US-DED-07	**Update Progress**	As a staff, I want to update progress percentage on my tasks so that stakeholders can see how much is done	**Medium**	
US-DED-08	**Receive Deadline Alerts**	As a staff, I want to receive automated deadline alerts so	**High**	

		that I don’t miss important dates		
US-DED-09	**Generate Compliance Reports**	As an admin, I want to generate deadlines compliance reports so that I can track team performance	**Medium**	
US-DED-10	**Filter Deadlines**	As a staff, I want to filter deadlines by status, date, range, and assignee so that I can focus on relevant tasks	**Medium**	
US-DED-11	**View Timeline**	As an admin, I want to view deadlines in a timeline format so that I can visualize task distribution over time	**Medium**	

US-DED-12	**Link Deadlines to Sessions**	As an admin, I want deadlines to automatically link to related sessions so that task context is clear	**High**	
US-DED-13	**Request Deadline Extension**	As a staff, I want to request deadline extensions with reasons so that Admin can approve or deny	**Low**	
MODULE 5: Reminder System				
US-REM-01	**Set Session Reminders**	As an admin, I want to set automatic reminders for sessions so that attendees are notified in advance	**High**	
US-REM-02	**Set Deadline Reminders**	As an admin, I want to set reminders for	**High**	

		deadlines so that staff are alerted before due dates		
US-REM-03	**Receive Email Reminders**	As a staff, I want to receive email reminders so that I stay informed even when not logged in	**Medium**	
US-REM-04	**Receive In-App Notifications**	As a staff, I want to receive in-app notifications so that I see reminders when logged into the system	**High**	
US-REM-05	**Customize Reminder Timing**	As an admin, I want to customize reminder timing (1days, 3days, 1 week before) so that notifications are sent at optimal times	**Medium**	

US-REM-06	**View Notification History**	As as staff, I want to view my notifications history so that I can see past reminders and alerts	**Medium**	
US-REM-07	**Mark as Read**	As a staff, I want to mark notifications as read so that I can keep track of what I've reviewed	**Medium**	
US-REM-08	**Snooze Reminders**	As a staff, I want to snooze reminders so that I can be reminder again later	**Low**	
US-REM-09	**View Reminder Logs**	As an admin, I want to view reminder logs so that I can confirm notifications were sent successfully	**Low**	

US-REM-10	**Configure Notification Channels**	As an admin, I want to configure notification channels (email, in-app, SMS) so that users receive alerts through preferred methods	**Medium**	
US-REM-11	**Set Quiet Hours**	As a staff, I want to set quiet hours for notifications so that I don’t receive alerts during off-hours	**Low**	
US-REM-12	**Dismiss Notifications**	As a staff, I want to dismiss notification so that my notification panel stays clean	**Medium**	
US-REM-13	**Receive Escalation Alerts**	As an admin, I want to receive escalations alerts for overdue items so that	**High**	

		I can take immediate action		
US-REM-14	**View Unread Count**	As a staff, I want to see an unread notification count badge so that I know how many alerts I need to review	**Medium**	

Table no. 3 Product Backlog

3.4.2 Product Backlog for EIS Information Security

EIS No.	EIS User Stories	EIS IS Priority	Revision Priority	Status
EIS-001	As a System Administrator, I want to implement role-based access control (RBAC) so that users can only access features appropriate to their role (Legislative Member, Committee Member, Administrative Staff, Public)	High	High	Pending
EIS-002	As a System Administrator, I want to enforce secure login with password requirements (minimum 8 characters, alphanumeric) so that unauthorized access is prevented.	High	High	Pending

EIS-003	As a System Administrator, I want to implement account lockout after 5 failed login attempts so that brute force attacks are prevented.	High	High	Pending
EIS-004	As a System Administrator, I want to implement session timeout after 30 minutes of inactivity so that unattended sessions cannot be exploited.	Medium	Medium	Pending
EIS-005	As a System Administrator, I want to log all login attempts (successful and failed) so that suspicious activities can be tracked.	Medium	Medium	Pending
EIS-006	As a System Administrator, I want all data transmitted between client and server to be encrypted using HTTPS/TLS so that data cannot be intercepted.	High	High	Pending
EIS-007	As a System Administrator, I want sensitive data in the database (passwords, personal information) to be encrypted so that data breaches do not expose readable information.	High	High	Pending
EIS-008	As a System Administrator, I want to implement regular database backups (daily) so that data can be recovered in case of system failure.	High	High	Pending
EIS-009	As a System Administrator, I want to log all critical user actions (create, update, delete on agendas, sessions, deadlines) with timestamp and user ID so that activities can be audited.	High	High	Pending

EIS-010	As a System Administrator, I want audit logs to be retained for at least 1 year so that historical investigations can be conducted.	Medium	Medium	Pending
EIS-011	As a Compliance Officer, I want to generate audit reports for specific date ranges and users so that compliance reviews can be performed.	Medium	Medium	Pending
EIS-012	As a System Administrator, I want to implement input validation on all forms so that SQL injection and XSS attacks are prevented.	High	High	Pending
EIS-013	As a Legislative Member, I want to receive notifications when agenda items or sessions I created are modified by others so that unauthorized changes are identified.	Medium	Medium	Pending
EIS-014	As a System Administrator, I want to maintain version history for agenda items so that changes can be tracked and reversed if needed.	Medium	Medium	Pending
EIS-015	As a System Administrator, I want CALENDAR MANAGEMENT to be viewable by all authenticated users but only editable by administrators so that calendar integrity is maintained.	High	High	Pending
EIS-016	As a System Administrator, I want SESSION SCHEDULING to be restricted to authorized schedulers and administrators only so that unauthorized sessions cannot be created.	High	High	Pending

EIS-017	As a System Administrator, I want AGENDA CREATION to allow members to propose items but require approval before publishing so that agenda quality is controlled.	High	High	Pending
EIS-018	As a System Administrator, I want voting records to be viewable only by authorized members and never editable after submission so that voting integrity is protected.	High	High	Pending
EIS-019	As a System Administrator, I want API connections with external systems to use secure authentication (API keys) so that unauthorized API access is prevented.	High	High	Pending
EIS-020	As a System Administrator, I want to validate all data received from external systems before processing so that malicious data cannot compromise the system.	Medium	Medium	Pending
EIS-021	As a System Administrator, I want to log all data exchanges with external systems so that integration issues can be investigated.	Low	Low	Pending
EIS-022	As a System Administrator, I want to restrict file uploads to approved formats only (PDF, DOCX, XLSX) so that malicious files cannot be uploaded.	High	High	Pending
EIS-023	As a System Administrator, I want to implement file size limits (maximum 25MB) so that system resources are not exhausted.	Medium	Medium	Pending
EIS-024	As a System Administrator, I want uploaded files to be	High	Medium	Pending

	scanned for viruses before storage so that malware cannot enter the system.			
EIS-025	As a System Administrator, I want automated database backups stored in a secure location so that data can be restored after incidents.	High	High	Pending
EIS-026	As a System Administrator, I want to test backup restoration quarterly so that recovery procedures are verified.	Medium	Medium	Pending
EIS-027	As a System Administrator, I want a documented incident response procedure so that security breaches can be handled systematically.	Medium	Low	Pending
EIS-028	As a System Administrator, I want to implement data retention policies that automatically archive old data after 5 years so that storage is optimized and compliance is maintained.	Medium	Low	Pending
EIS-029	As a Data Privacy Officer, I want users to be able to request access to their personal data so that privacy rights are respected.	Low	Low	Pending
EIS-030	As a System Administrator, I want to anonymize personal information in public reports so that privacy is protected.	Medium	Medium	Pending

Table no. 4 Product Backlog for EIS Information Security

3.4.3 Product Backlog for EIS Standards

3.4.3.1 UI/UX

EIS Standard No.	EIS Standard User Stories	EIS Standard Priority	Revision Priority	Status
EIS-031	As a User, I want the system to display the Valenzuela City logo in the header so that I can identify the system's jurisdiction.	High	High	Pending
EIS-032	As a User, I want the primary color scheme to use red tones (consistent with client requirements) with neutral complementary colors so that the system has a cohesive brand identity.	High	High	Pending
EIS-033	As a User, I want consistent spacing and typography using Tailwind CSS utility classes so that the interface is visually organized.	High	High	Pending
EIS-034	As a Mobile User, I want the calendar and agenda views to be responsive on tablet and mobile devices so that I can access information on different devices.	High	High	Pending
EIS-035	As a User, I want navigation menus to be mobile-friendly (collapsible) so that screen space is optimized.	Medium	High	Pending
EIS-036	As a User, I want a consistent navigation bar with clear labels for main modules (Calendar,	High	High	Pending

	Sessions, Agenda, Deadlines) so that I can easily navigate the system.			
EIS-037	s a User, I want the current active page to be visually highlighted in the navigation so that I know where I am in the system.	Medium	Medium	Pending
EIS-038	As a User, I want the calendar to display in month view with easy navigation so that I can view schedules clearly.	High	High	Pending
EIS-039	As a User, I want different event types (sessions, deadlines, hearings) to be color-coded on the calendar so that I can quickly distinguish between activities.	High	High	Pending
EIS-040	As a User, I want today's date to be highlighted on the calendar so that I maintain temporal awareness.	Medium	Medium	Pending
EIS-041	As a Scheduler, I want a clear form with labeled fields (date, time, location) for creating sessions so that I can input information accurately.	High	High	Pending
EIS-042	As a User, I want session items to display key information (title, date, time, status) in a readable format so that I can quickly review scheduled sessions.	High	High	Pending
EIS-043	As a User, I want visual indicators (badges) to	Medium	Medium	Pending

	show session status (upcoming, completed, cancelled) so that I understand session states at a glance.			
EIS-044	As a Legislative Member, I want a text editor for agenda item descriptions so that I can format content appropriately.	High	High	Pending
EIS-045	As a User, I want agenda items to be displayed in an ordered list so that the sequence is clear.	High	High	Pending
EIS-046	As a User, I want visual indicators for agenda item status (draft, approved, completed) using badges so that I can track progress	Medium	Medium	Pending
EIS-047	As a User, I want overdue deadlines to be highlighted in red so that urgent items are immediately visible.	High	High	Pending
EIS-048	As a User, I want upcoming deadlines to be highlighted with a warning color so that I have advance notice.	High	High	Pending
EIS-049	As a User, I want a list showing my upcoming deadlines sorted by date so that I can prioritize my work.	High	High	Pending
EIS-050	As a User, I want reminder notifications to appear as toast messages so that I'm alerted without disrupting my workflow.	High	High	Pending

EIS-051	As a User, I want a notification icon with a badge showing unread reminder count so that I can see pending notifications.	Medium	High	Pending
EIS-052	As a User, I want all required form fields to be clearly marked with an asterisk (*) so that I know what information is mandatory.	High	High	Pending
EIS-053	As a User, I want clear error messages displayed below fields when validation fails so that I can correct mistakes immediately.	High	High	Pending
EIS-054	As a User, I want success messages after successfully submitting forms so that I have confirmation of my actions.	Medium	Medium	Pending
EIS-055	As a User, I want primary action buttons (Save, Submit, Create) to be styled prominently in red so that key actions are emphasized.	High	High	Pending
EIS-056	s a User, I want secondary action buttons (Cancel, Back) to have neutral styling so that they're distinguished from primary actions.	High	High	Pending
EIS-057	As a User, I want destructive actions (Delete) to require confirmation	High	High	Pending

	dialogs so that accidental deletions are prevented.			
EIS-058	As a User, I want tables to have clear column headers so that data is organized and understandable.	High	High	Pending
EIS-059	As a User, I want sortable columns (with arrow indicators) so that I can organize data by different criteria.	Medium	High	Pending
EIS-060	As a User, I want pagination controls on long tables so that performance is optimized.	Medium	Medium	Pending
EIS-061	As a User, I want a search bar with clear placeholder text so that search functionality is discoverable.	High	High	Pending
EIS-062	As a User, I want basic filter options (dropdown or checkboxes) so that I can narrow down results.	Medium	Medium	Pending
EIS-063	As a User, I want modals to have a backdrop that dims the background so that focus is on the dialog content.	High	High	Pending
EIS-064	As a User, I want modals to be closeable via an X button or Cancel button so that I can exit easily.	High	High	Pending
EIS-065	As a User, I want loading spinners to appear during data fetching so that I know the system is working.	High	High	Pending
EIS-066	As a User, I want success messages to appear as	High	High	Pending

	green notifications so that I know my actions completed successfully.			
EIS-067	As a User, I want error messages to appear as red notifications so that I'm aware of problems.	High	High	Pending
EIS-068	As a User, I want sufficient color contrast on all text so that content is readable.	High	High	Pending
EIS-069	As a User, I want all interactive elements to be keyboard accessible so that I can navigate without a mouse.	Medium	High	Pending
EIS-070	As a User, I want a dashboard that displays key metrics (upcoming sessions, pending agenda items, overdue deadlines) so that I can see important information at a glance.	High	High	Pending
EIS-071	As a User, I want a calendar widget on the dashboard showing this month's activities so that I have quick access to schedules.	Medium	High	Pending
EIS-072	As a User, I want the system to use a clear, readable font so that text is easy to read.	High	High	Pending
EIS-073	As a User, I want consistent font sizing for headings and body text so that content structure is clear.	Medium	Medium	Pending
EIS-074	As a User, I want consistent icons throughout	Medium	High	Pending

	the system so that visual language is unified.			
EIS-075	As a User, I want status icons (checkmarks, warnings) to use clear symbols so that I can quickly interpret states.	Medium	Medium	Pending
EIS-076	As a User, I want empty state screens to include helpful guidance and action buttons so that I know how to get started.			

Table no. 5 Product Backlog for EIS Standards

3.4.4 Product Backlog for EIS Integration

EIS Integration No.	EIS Integration User Stories	EIS Integration Priority	Revision Priority	Status
EIS-077	As a System, I want CALENDAR MANAGEMENT to receive tasks and deadlines from DEADLINE TRACKING AND MONITORING so that all deadlines are displayed on the calendar.	High	High	Pending
EIS-078	As a System, I want CALENDAR MANAGEMENT to reflect session schedules from SESSION SCHEDULING	High	High	Pending

	(bidirectional sync) so that calendar events stay synchronized with session data.			
EIS-079	As a System, I want CALENDAR MANAGEMENT to reflect agendas from AGENDA CREATION AND MANAGEMENT (bidirectional sync) so that agenda items appear on the calendar.	High	High	Pending
EIS-080	As a System, I want CALENDAR MANAGEMENT to receive notification triggers from REMINDER SYSTEM so that reminders are linked to calendar events.	Medium	High	Pending
EIS-081	As a System, I want SESSION SCHEDULING to provide sessions to AGENDA CREATION AND MANAGEMENT and receive agenda items (bidirectional sync) so that sessions and agendas are coordinated.	High	High	Pending
EIS-082	As a System, I want SESSION SCHEDULING to give session schedules to DEADLINE TRACKING AND	High	High	Pending

	MONITORING (bidirectional sync) so that session deadlines are tracked.			
EIS-083	As a System, I want SESSION SCHEDULING to provide session schedules to CALENDAR MANAGEMENT so that sessions appear on the calendar.	High	High	Pending
EIS-084	As a System, I want SESSION SCHEDULING to provide sessions to REMINDER SYSTEM so that session reminders can be generated.	Medium	High	Pending
EIS-085	As a System, I want AGENDA CREATION AND MANAGEMENT to provide agenda items to SESSION SCHEDULING and receive sessions (bidirectional sync) so that agendas are linked to sessions.	High	High	Pending
EIS-086	As a System, I want AGENDA CREATION AND MANAGEMENT to reflect agenda items to CALENDAR MANAGEMENT (bidirectional sync) so that agenda items	High	High	Pending

	appear on the calendar.			
EIS-087	As a System, I want AGENDA CREATION AND MANAGEMENT to give agenda item deadlines to DEADLINE TRACKING AND MONITORING and be notified when agendas are completed (bidirectional sync) so that agenda progress is tracked.	High	High	Pending
EIS-090		High	High	Pending
EIS-091	As a System, I want DEADLINE TRACKING AND MONITORING to give notifications to REMINDER SYSTEM for tasks that need to be done so that reminders are triggered.	Medium	High	Pending
EIS-092	As a System, I want REMINDER SYSTEM to receive sessions from SESSION SCHEDULING so that session reminders can be created.	Medium	High	Pending
EIS-093	As a System, I want REMINDER SYSTEM to receive agenda items from AGENDA CREATION AND MANAGEMENT so	Medium	High	Pending

	that agenda reminders can be created.			
EIS-094	As a System, I want REMINDER SYSTEM to receive notifications of deadlines from DEADLINE TRACKING AND MONITORING so that deadline reminders can be created.	Medium	High	Pending
EIS-095	As a System, I want REMINDER SYSTEM to give notification triggers to CALENDAR MANAGEMENT so that reminders are displayed on the calendar.	Medium	Medium	Pending
EIS-096	As a System, I want SESSION SCHEDULING to give session schedules to DOCUMENT STORAGE & VERSIONING (bidirectional sync) so that document preparation timelines are coordinated.	High	High	Pending
EIS-097	As a System, I want SESSION SCHEDULING to receive legislative trend reports from LEGISLATIVE TREND ANALYSIS & TAGGING so that	Medium	Medium	Pending

	session planning is informed by trends.			
EIS-098	As a System, I want AGENDA CREATION AND MANAGEMENT to receive relevant documents from DOCUMENT STORAGE & VERSIONING (bidirectional sync) so that documents can be attached to agenda items.	High	High	Pending
EIS-099	As a System, I want AGENDA CREATION AND MANAGEMENT to pull comparative analysis results from ADVANCED SEARCH, DISCOVERY & COMPARATIVE DOCUMENT ANALYSIS so that agenda items have supporting analysis.	Medium	Medium	Pending
EIS-100	As a System, I want CALENDAR MANAGEMENT to display public hearing schedules from HEARING MANAGEMENT/PUBLIC LIVE STREAM (bidirectional sync) so that hearings appear on the calendar.	High	High	Pending
EIS-101	As a System, I want SESSION	High	High	Pending

	SCHEDULING to coordinate with SCHEDULING MANAGEMENT (bidirectional sync) so that conflicts between sessions and public hearings are avoided.			
EIS-102	As a System, I want SESSION SCHEDULING to provide session schedules to HEARING MANAGEMENT/PUBLIC LIVE STREAM (bidirectional sync) so that hearing integration is planned.	Medium	High	Pending
EIS-103	As a System, I want AGENDA CREATION AND MANAGEMENT to send agenda items requiring public input to HEARING MANAGEMENT/PUBLIC LIVE STREAM so that public hearings are scheduled for relevant items.	Medium	High	Pending
EIS-104	As a System, I want AGENDA CREATION AND MANAGEMENT to receive public hearing topics from SCHEDULING MANAGEMENT so that hearing topics are included in agendas.	Medium	Medium	Pending

EIS-105	As a System, I want SESSION SCHEDULING to receive committee availability from COMMITTEE PROFILES & MEMBERSHIP so that scheduling is coordinated with committee availability.	High	High	Pending
EIS-106	As a System, I want SESSION SCHEDULING to coordinate session dates with COMMITTEE MEETINGS MANAGEMENT (bidirectional sync) so that session and committee meeting overlaps are avoided.	High	High	Pending
EIS-107	As a System, I want AGENDA CREATION AND MANAGEMENT to send legislation to REFERRAL TRACKING & HANDLING for committee review so that items are routed to appropriate committees.	High	High	Pending
EIS-108	As a System, I want AGENDA CREATION AND MANAGEMENT to pull deliberation topics from AGENDA & DELIBERATION	Medium	Medium	Pending

	MANAGEMENT so that committee deliberations are included in main session agendas.			
EIS-109	As a System, I want AGENDA CREATION AND MANAGEMENT to coordinate agenda items with ACTION ITEMS & FOLLOW-UPS (bidirectional sync) so that committee outputs are tracked.	Medium	Medium	Pending
EIS-110	As a System, I want SESSION SCHEDULING to provide session dates to READING & WORKFLOW TRACKING so that first, second, and third reading schedules are coordinated.	High	High	Pending
EIS-111	As a System, I want SESSION SCHEDULING to coordinate with VOTING & ENACTMENT (bidirectional sync) so that voting session scheduling is synchronized.	High	High	Pending
EIS-112	As a System, I want AGENDA CREATION AND MANAGEMENT to send approved agenda items to	High	High	Pending

	DOCUMENT FINALIZATION so that items are officially documented.			
EIS-113	As a System, I want AGENDA CREATION AND MANAGEMENT to receive document status updates from DOCUMENT STATUS & LOCATION so that agenda item progress is updated.	Medium	High	Pending
EIS-114	As a System, I want SESSION SCHEDULING to provide session schedules to CONSULTATION MANAGEMENT and receive consultation period dates (bidirectional sync) so that consultation timing is coordinated with sessions.	Medium	High	Pending
EIS-115	As a System, I want SESSION SCHEDULING to provide session records to ARCHIVE INTAKE & CLASSIFICATION so that session history is preserved.	Medium	Medium	Pending
EIS-116	As a System, I want AGENDA CREATION AND MANAGEMENT to send finalized agendas to ARCHIVE	Medium	Medium	Pending

	INTAKE & CLASSIFICATION so that agenda history is permanently stored.			
EIS-117	As a System, I want CALENDAR MANAGEMENT to display voting schedules from VOTING SESSIONS MANAGEMENT (bidirectional sync) so that voting events appear on the calendar.	High	High	Pending
EIS-118	As a System, I want SESSION SCHEDULING to provide plenary session schedules to VOTING SESSIONS MANAGEMENT so that voting is coordinated with sessions.	High	High	Pending
EIS-119	As a System, I want DEADLINE TRACKING AND MONITORING to receive voting deadlines from VOTING SESSIONS MANAGEMENT and alert when voting is scheduled (bidirectional sync) so that voting deadlines are tracked.	High	High	Pending
EIS-120	As a System, I want SESSION	High	High	Pending

	SCHEDULING to coordinate directly with MEETING COORDINATION AND SCHEDULING (bidirectional sync) so that unified session planning is achieved.			
EIS-121	As a System, I want AGENDA CREATION AND MANAGEMENT to send session agendas to MEETING COORDINATION AND SCHEDULING so that meeting preparation includes agenda information.	High	High	Pending
EIS-122	As a Developer, I want all module integrations to use RESTful API endpoints with standard HTTP methods (GET, POST, PUT, DELETE) so that integration is standardized and maintainable.	High	High	Pending
EIS-123	As a Developer, I want API responses to follow a consistent JSON format with status codes, data, and error messages so that error handling is uniform across integrations.	High	High	Pending

EIS-124	As a Developer, I want all API integrations to implement proper error handling and retry logic so that temporary failures do not break data synchronization.	High	High	Pending
EIS-125	As a Developer, I want API endpoints to include versioning (e.g., /api/v1/) so that future updates do not break existing integrations.	Medium	High	Pending
EIS-126	As a System Administrator, I want bidirectional data synchronization to include conflict resolution logic so that data inconsistencies are handled appropriately.	High	High	Pending
EIS-127	As a System Administrator, I want all data exchanges to be logged with timestamps and user identification so that integration activities can be audited.	Medium	High	Pending
EIS-128	As a System Administrator, I want failed integration attempts to trigger alerts to system administrators so that	Medium	Medium	Pending

	issues are promptly addressed.			
EIS-129	As a QA Tester, I want integration test cases to verify data flow between all connected modules so that integration functionality is validated.	High	High	Pending
EIS-130	As a QA Tester, I want integration tests to include scenarios for both successful and failed data exchanges so that error handling is verified.	Medium	Medium	Pending

Table no. 6 Product Backlog for EIS Integration

3.4.5 Product Backlog for Analytics

3.4.5.1 Application System Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-1	As a Legislative Member, I want to view the total number of scheduled sessions for the current month so that I can understand the workload.	High	High	Pending

ASA-2	As a Legislative Member, I want to view the total number of pending agenda items so that I can prioritize my work.	High	High	Pending
ASA-3	As a Legislative Member, I want to view the total number of overdue deadlines so that I can identify urgent tasks.	High	High	Pending
ASA-4	As a Legislative Member, I want to view the total number of completed tasks for the current month so that I can track productivity.	Medium	Medium	Pending
ASA-5	As an Administrator, I want to view key performance indicators (KPIs) on the dashboard (sessions completed, agenda items approved, deadlines met) so that I can monitor overall performance.	Medium	High	Pending

Table no. 8 Product Backlog for Dashboard Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status

ASA-6	As an Administrator, I want to view the total number of sessions conducted per month so that I can analyze session frequency trends.	High	High	Pending
ASA-7	As an Administrator, I want to view session completion rates (completed vs scheduled) so that I can identify scheduling issues.	High	High	Pending
ASA-8	As an Administrator, I want to view average session duration so that I can optimize scheduling.	Medium	Medium	Pending
ASA-9	As an Administrator, I want to view sessions by type (plenary, committee, special) so that I can analyze session distribution.	Medium	Medium	Pending

Table no. 9 Product Backlog for Session Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-10	As a Legislative Member, I want to view the total number	High	High	Pending

	of agenda items created per month so that I can track legislative activity.			
ASA-11	As a Legislative Member, I want to view agenda item status distribution (draft, pending approval, approved, completed) so that I can understand the workflow pipeline.	High	High	Pending
ASA-12	As an Administrator, I want to view average time from agenda item creation to approval so that I can identify bottlenecks.	Medium	High	Pending
ASA-13	As an Administrator, I want to view agenda items by category or type so that I can analyze legislative priorities.	Medium	Medium	Pending
ASA-14	As a Legislative Member, I want to view the number of agenda items I have authored so that I can track my contributions.	Low	Medium	Pending

Table no. 10 Product Backlog for Agenda Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-15	As a Legislative Member, I want to view deadline compliance rate (met vs missed) so that I can assess time management.	High	High	Pending
ASA-16	As an Administrator, I want to view the total number of overdue deadlines by user so that I can identify individuals needing support.	High	High	Pending
ASA-17	As an Administrator, I want to view average days to deadline completion so that I can analyze efficiency.	Medium	Medium	Pending
ASA-18	As a Legislative Member, I want to view upcoming deadlines for the next 7 days so that I can plan my work.	High	High	Pending

Table no. 11 Product Backlog for Deadline Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-19	As an Administrator, I want to view the total number of reminders sent per month so that I	Medium	Medium	Pending

	can monitor system notification activity.			
ASA-20	As an Administrator, I want to view reminder response rate (acted upon vs ignored) so that I can assess reminder effectiveness.	Medium	Low	Pending
ASA-21	As a Legislative Member, I want to view my most frequent reminder types so that I can identify recurring tasks.	Low	Low	Pending

Table no. 12 Product Backlog for Reminder Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-22	As an Administrator, I want to view calendar occupancy rate (scheduled days vs available days) so that I can optimize scheduling.	Medium	High	Pending
ASA-23	As an Administrator, I want to view peak scheduling periods (days/months with most activities) so that I can plan resource allocation.	Medium	Medium	Pending

ASA-24	As an Administrator, I want to view scheduling conflicts (overlapping sessions or hearings) so that I can improve coordination.	High	High	Pending

Table no. 13 Product Backlog for Calendar Utilization Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-25	As an Administrator, I want to view the most active users (by sessions created, agendas authored, tasks completed) so that I can recognize contributions.	Medium	Medium	Pending
ASA-26	As an Administrator, I want to view user login frequency so that I can assess system engagement.	Low	Low	Pending
ASA-27	As an Administrator, I want to view average response time to reminders by user so that I can identify engagement levels	Low	Low	Pending

Table no. 14 Product Backlog for User Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-28	As an Administrator, I want to view the number of agenda items sent to committees so that I can monitor referral activity.	Medium	Medium	Pending
ASA-29	As an Administrator, I want to view the number of sessions linked to public hearings so that I can assess public engagement.	Medium	Medium	Pending
ASA-30	As an Administrator, I want to view document attachment rate (agendas with documents vs without) so that I can ensure proper documentation.	Low	Medium	Pending

Table no. 15 Product Backlog for Integration Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-31	As an Administrator, I want to generate monthly summary reports (sessions, agendas, deadlines) in PDF format so that I can share performance data.	High	High	Pending

ASA-32	As an Administrator, I want to export analytics data to Excel/CSV format so that I can perform custom analysis.	Medium	High	Pending
ASA-33	As an Administrator, I want to schedule automatic monthly reports to be generated and emailed so that reporting is automated.	Low	Medium	Pending

Table no. 15 Product Backlog for Report and Export Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-34	As a User, I want to view session trends in a line chart (monthly over time) so that I can visualize activity patterns.	Medium	High	Pending
ASA-35	As a User, I want to view agenda status distribution in a pie chart so that I can quickly understand the workflow state.	Medium	High	Pending
ASA-36	As a User, I want to view deadline compliance in a bar chart (met vs	Medium	Medium	Pending

	missed by month) so that I can track performance visually.			
ASA-37	As an Administrator, I want to view calendar heatmap showing activity density so that I can identify busy periods.	Low	Medium	Pending

Table no. 15 Product Backlog for Vizualization Analytics

EIS No.	EIS Analytics User Stories	EIS Priority	Revision Priority	Status
ASA-38	As a User, I want to filter all analytics by date range (custom, this month, this quarter, this year) so that I can analyze specific periods.	High	High	Pending
ASA-39	As an Administrator, I want to filter analytics by user or department so that I can analyze individual or team performance.	Medium	Medium	Pending
ASA-40	As an Administrator, I want to drill down from summary metrics to detailed records so	Low	Medium	Pending

	that I can investigate specific cases.			

Table no. 15 Product Backlog for Filtering and Drill-down Analytics

3.4.6 Sprint Backlog

SPRINT 1: FOUNDATION AND SECURITY (Weeks 1-2)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-001	EIS-001	As a System Administrator, I want to implement role-based access control (RBAC) so that users can only access features appropriate to their role.	Design database schema for roles and permissions; Implement RBAC middleware; Create role assignment interface	Week 1-2	Lead Programmer, Programmer

T-002	EIS-002	As a System Administrator, I want to enforce secure login with password requirements so that unauthorized access is prevented.	Implement password validation logic; Create password policy configuration; Design login form with validation	Week 1	Programmer
T-003	EIS-003	As a System Administrator, I want to implement account lockout after 5 failed login attempts so that brute force attacks are prevented.	Implement login attempt counter; Create lockout mechanism; Design lockout notification	Week 1	Programmer
T-004	EIS-006	As a System Administrator, I want all data transmitted between client and server to be encrypted using HTTPS/TLS so that data cannot be intercepted.	Configure SSL/TLS certificates; Implement HTTPS enforcement; Test secure connections	Week 1	Lead Programmer
T-005	EIS-007	As a System Administrator, I want sensitive data in the database to be encrypted so	Implement database encryption for passwords; Set up encryption keys	Week 2	Lead Programmer

		that data breaches do not expose readable information	management; Test data encryption		
T-006	EIS-012	As a System Administrator, I want to implement input validation on all forms so that SQL injection and XSS attacks are prevented.	Create validation middleware; Implement sanitization functions; Test against common attacks	Week 2	Lead Programmer, Programmer
T-007	N/A	Project documentation - Security standards	Document security implementation; Create security policy documentation; Document encryption methods	Week 1-2	Documentation Specialist
T-008	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Track sprint progress	Week 1-2	Project Manager

SPRINT 2: UI/UX FOUNDATION AND BRANDING (Weeks 3-4)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-009	EIS-031	As a User, I want the system to display the Valenzuela City logo in the header so that I can identify the system's jurisdiction.	Design header layout; Integrate Valenzuela City logo; Implement responsive header	Week 3	Programmer
T-010	EIS-032	As a User, I want the primary color scheme to use red tones with neutral complementary colors so that the system has a cohesive brand identity	Create Tailwind color configuration; Design color palette; Implement theme across components	Week 3	Lead Programmer, Programmer
T-011	EIS-033	As a User, I want consistent spacing and typography using Tailwind CSS so that the interface is visually organized.	Configure Tailwind CSS; Set up typography scale; Create spacing utilities	Week 3	Programmer
T-012	EIS-036	As a User, I want a consistent navigation bar with clear labels	Design navigation component; Implement navigation menu;	Week 3-4	Programmer

		for main modules so that I can easily navigate the system.	Create navigation logic		
T-013	EIS-037	As a User, I want the current active page to be visually highlighted in the navigation so that I know where I am in the system.	Implement active state tracking; Apply active styling; Test navigation states	Week 4	Programmer
T-014	EIS-068	As a User, I want sufficient color contrast on all text so that content is readable.	Audit color contrast ratios; Adjust colors for WCAG compliance; Document contrast standards	Week 4	Programmer, Documentation Specialist
T-015	EIS-072	As a User, I want the system to use a clear, readable font so that text is easy to read.	Select and configure font family; Implement font loading; Test readability	Week 3	Programmer
T-016	N/A	Project documentation - UI/UX standards	Document UI component library; Create style guide; Document Tailwind configuratio	Week 3-4	Documentation Specialist

T-017	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review UI/UX implementation	Week 3-4	Project Manager

SPRINT 3: RESPONSIVE DESIGN AND CORE COMPONENTS (Weeks 5-6)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-018	EIS-034	As a Mobile User, I want the calendar and agenda views to be responsive on tablet and mobile devices so that I can access information on different devices.	Design responsive breakpoints; Implement mobile layouts; Test on multiple devices	Week 5-6	Lead Programmer, Programmer
T-019	EIS-035	As a User, I want navigation	Create hamburger menu	Week 5	Programmer

		menus to be mobile-friendly so that screen space is optimized.	component; Implement mobile navigation; Add touch interactions		
T-020	EIS-055	As a User, I want primary action buttons to be styled prominently in red so that key actions are emphasized.	Design button component system; Implement button variants; Create button documentation	Week 5	Programmer
T-021	EIS-056	As a User, I want secondary action buttons to have neutral styling so that they're distinguished from primary actions.	Implement secondary button styles; Test button hierarchy; Document button usage	Week 5	Programmer
T-022	EIS-063	As a User, I want modals to have a backdrop that dims the background so that focus is on the dialog content.	Create modal component; Implement backdrop overlay; Add animation transitions	Week 6	Programmer
T-023	EIS-064	As a User, I want modals to be closeable via an X button or Cancel	Implement modal close functionality; Add keyboard support (ESC); Test	Week 6	Programmer

		button so that I can exit easily.	modal interactions		
T-024	EIS-069	As a User, I want all interactive elements to be keyboard accessible so that I can navigate without a mouse.	Implement tab navigation; Add focus indicators; Test keyboard accessibility	Week 6	Programmer
T-025	N/A	Project documentation - Responsive design	Document responsive breakpoints; Create mobile testing checklist; Document component responsiveness	Week 5-6	Documentation Specialist
T-026	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review responsive implementati	Week 5-6	Project Manage

SPRINT 4: CALENDAR MANAGEMENT MODULE (Weeks 7-8)

Task No.	**User Story No.**	**User Stories**	**Tasks**	**Timeline**	**Responsible Team Member/s**

T-027	EIS-038	As a User, I want the calendar to display in month view with easy navigation so that I can view schedules clearly.	Design calendar grid layout; Implement month view logic; Add navigation controls	Week 7	Lead Programmer, Programmer
T-028	EIS-039	As a User, I want different event types to be color-coded on the calendar so that I can quickly distinguish between activities.	Create event type color mapping; Implement color-coded events; Add legend component	Week 7	Programmer
T-029	EIS-040	As a User, I want today's date to be highlighted on the calendar so that I maintain temporal awareness.	Implement current date detection; Apply highlight styling; Test date highlighting	Week 7	Programmer
T-030	EIS-015	As a System Administrator, I want CALENDAR MANAGEMENT to be viewable by all authenticated users but only editable by administrators so that calendar integrity is maintained.	Implement view permissions; Add edit restrictions; Create permission checks	Week 8	Lead Programmer

T-031	EIS-077	As a System, I want CALENDAR MANAGEMENT to receive tasks and deadlines from DEADLINE TRACKING so that all deadlines are displayed on the calendar.	Design API endpoint for deadline sync; Implement data integration; Test deadline display	Week 8	Lead Programmer
T-032	EIS-078	As a System, I want CALENDAR MANAGEMENT to reflect session schedules from SESSION SCHEDULING so that calendar events stay synchronized.	Design bidirectional sync mechanism; Implement session-calendar sync; Test synchronization	Week 8	Lead Programmer,
T-033	N/A	Project documentation - Calendar module	Document calendar API; Create user guide for calendar; Document color coding system	Week 7-8	Documentation Specialist
T-034	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review calendar implementation	Week 7-8	Project Manager

SPRINT 5: SESSION SCHEDULING MODULE (Weeks 9-10)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-035	EIS-041	As a Scheduler, I want a clear form with labeled fields for creating sessions so that I can input information accurately.	Design session creation form; Implement form fields; Add validation	Week 9	Programmer
T-036	EIS-042	As a User, I want session items to display key information in a readable format so that I can quickly review scheduled sessions.	Design session card/list component; Implement session display; Add sorting functionality	Week 9	Programmer
T-037	EIS-043	As a User, I want visual indicators to show session status so that I understand session states at a glance.	Create status badge component; Implement status logic; Apply status styling	Week 9	Programmer

T-038	EIS-016	As a System Administrator, I want SESSION SCHEDULING to be restricted to authorized schedulers and administrators only so that unauthorized sessions cannot be created.	Implement role-based creation permissions; Add authorization checks; Test permission enforcement	Week 10	Lead Programmer
T-039	EIS-081	As a System, I want SESSION SCHEDULING to provide sessions to AGENDA CREATION and receive agenda items so that sessions and agendas are coordinated.	Design session-agenda API integration; Implement bidirectional sync; Test data flow	Week 10	Lead Programmer
T-040	EIS-083	As a System, I want SESSION SCHEDULING to provide session schedules to CALENDAR MANAGEMENT so that sessions	Implement session-to-calendar sync; Test calendar updates; Validate data integrity	Week 10	Lead Programmer

		appear on the calendar.			
T-041	N/A	Project documentation - Session scheduling	Document session scheduling API; Create scheduler user guide; Document session workflow	Week 9-10	Documentation Specialist
T-042	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review session module	Week 9-10	Project Manager

SPRINT 6: AGENDA CREATION AND MANAGEMENT MODULE (Weeks 11-12)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-043	EIS-044	As a Legislative Member, I want a text editor for agenda item descriptions so that I can format content appropriately.	Integrate rich text editor; Configure formatting options; Test editor functionality	Week 11	Programmer
T-044	EIS-045	As a User, I want agenda	Design agenda list component;	Week 11	Programmer

		items to be displayed in an ordered list so that the sequence is clear.	Implement ordering logic; Add reorder functionality		
T-045	EIS-046	As a User, I want visual indicators for agenda item status using badges so that I can track progress.	Create agenda status badges; Implement status workflow; Apply status colors	Week 11	Programmer
T-046	EIS-017	As a System Administrator, I want AGENDA CREATION to allow members to propose items but require approval before publishing so that agenda quality is controlled.	Implement approval workflow; Create approval interface; Add notification system	Week 12	Lead Programmer, Programmer
T-047	EIS-014	As a System Administrator, I want to maintain version history for agenda items so that changes can be tracked and	Design version control system; Implement version tracking; Create version comparison view	Week 12	Lead Programmer

		reversed if needed.			
T-048	EIS-085	As a System, I want AGENDA CREATION to provide agenda items to SESSION SCHEDULING and receive sessions so that agendas are linked to sessions.	Implement agenda-session integration; Test bidirectional sync; Validate relationships	Week 12	Lead Programmer
T-049	N/A	Project documentation - Agenda management	Document agenda API; Create legislative member guide; Document approval workflow	Week 11-12	Documentation Specialist
T-050	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review agenda module	Week 11-12	Project Manager

SPRINT 7: DEADLINE TRACKING MODULE (Weeks 13-14)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s

T-051	EIS-047	As a User, I want overdue deadlines to be highlighted in red so that urgent items are immediately visible.	Implement deadline status logic; Apply overdue styling; Create visual indicators	Week 13	Programmer
T-052	EIS-048	As a User, I want upcoming deadlines to be highlighted with a warning color so that I have advance notice.	Implement deadline proximity detection; Apply warning styling; Test highlighting logic	Week 13	Programmer
T-053	EIS-049	As a User, I want a list showing my upcoming deadlines sorted by date so that I can prioritize my work.	Design deadline list component; Implement sorting functionality; Add filtering options	Week 13	Programmer
T-054	EIS-088	As a System, I want DEADLINE TRACKING to notify CALENDAR MANAGEMENT when tasks are completed so that calendar	Design completion notification system; Implement status update API; Test calendar synchronization	Week 14	Lead Programmer

		events are updated.			
T-055	EIS-089	As a System, I want DEADLINE TRACKING to receive session schedules from SESSION SCHEDULING and notify when sessions are completed so that session completion is tracked.	Implement session-deadline integration; Create completion tracking; Test notification flow	Week 14	Lead Programmer
T-056	EIS-090	As a System, I want DEADLINE TRACKING to receive agenda items and notify when items are completed so that agenda completion is tracked.	Implement agenda-deadline integration; Create completion workflow; Test tracking accuracy	Week 14	Lead Programmer
T-057	N/A	Project documentation - Deadline tracking	Document deadline API; Create user guide for deadlines; Document notification system	Week 13-14	Documentation Specialist

T-058	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review deadline module	Week 13-14	Project Manager

SPRINT 8: REMINDER SYSTEM MODULE (Weeks 15-16)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-059	EIS-050	As a User, I want reminder notifications to appear as toast messages so that I'm alerted without disrupting my workflow.	Design toast notification component; Implement notification system; Add auto-dismiss functionality		Programmer
T-060	EIS-051	As a User, I want a notification icon with a badge showing unread reminder count so that I can see pending notifications.	Create notification badge component; Implement unread counter; Add click-to-view functionality		Programmer
T-061	EIS-092	As a System, I want REMINDER	Design session reminder integration;		Lead Programmer

		SYSTEM to receive sessions from SESSION SCHEDULING so that session reminders can be created.	Implement reminder creation logic; Test reminder triggers		
T-062	EIS-093	As a System, I want REMINDER SYSTEM to receive agenda items from AGENDA CREATION so that agenda reminders can be created.	Design agenda reminder integration; Implement reminder scheduling; Test reminder delivery		Lead Programmer
T-063	EIS-094	As a System, I want REMINDER SYSTEM to receive notifications of deadlines from DEADLINE TRACKING so that deadline reminders can be created.	Design deadline reminder integration; Implement reminder timing; Test notification accuracy		Lead Programmer
T-064	EIS-004	As a System Administrator, I want to implement session timeout after 30	Implement session timeout logic; Add timeout warning; Create session		Lead Programmer

		minutes of inactivity so that unattended sessions cannot be exploited.	refresh mechanism		
T-065	N/A	Project documentation - Reminder system	Document reminder API; Create user guide for notifications; Document reminder configuration	Week 15-16	Documentation Specialist
T-066	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review reminder implementation	Week 15-16	Project Manager

SPRINT 9: FORMS, TABLES, AND DATA DISPLAY (Weeks 17-18)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-067	EIS-052	As a User, I want all required form fields to be clearly marked with an asterisk so that I know what	Implement required field indicators; Update all forms; Create form validation guide	Week 17	Programmer

		information is mandatory.			
T-068	EIS-053	As a User, I want clear error messages displayed below fields when validation fails so that I can correct mistakes immediately.	Design error message component; Implement inline validation; Create error message library	Week 17	Programmer
T-069	EIS-054	As a User, I want success messages after successfully submitting forms so that I have confirmation of my actions.	Implement success notification system; Add form submission feedback; Test user experience	Week 17	Programmer
T-070	EIS-058	As a User, I want tables to have clear column headers so that data is organized and understandable.	Design table component; Implement column headers; Add responsive table layout	Week 18	Programmer
T-071	EIS-059	As a User, I want sortable columns so that I can organize data	Implement column sorting logic; Add sort indicators; Test sorting functionality	Week 18	Programmer

		by different criteria.			
T-072	EIS-060	As a User, I want pagination controls on long tables so that performance is optimized.	Design pagination component; Implement pagination logic; Add page size options	Week 18	Programmer
T-073	EIS-057	As a User, I want destructive actions to require confirmation dialogs so that accidental deletions are prevented.	Create confirmation dialog component; Implement confirmation workflow; Test deletion safety	Week 18	Programmer
T-074	N/A	Project documentation - Forms and tables	Document form validation rules; Create table component guide; Document data display standards	Week 17-18	Documentation Specialist
T-075	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review forms and tables	Week 17-18	Project Manager

SPRINT 10: SEARCH, FILTERING, AND LOADING STATES (Weeks 19-20)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-076	EIS-061	As a User, I want a search bar with clear placeholder text so that search functionality is discoverable.	Design search component; Implement search functionality; Add search suggestions	Week 19	Programmer
T-077	EIS-062	As a User, I want basic filter options so that I can narrow down results.	Design filter component; Implement filtering logic; Add filter persistence	Week 19	Programmer
T-078	EIS-065	As a User, I want loading spinners to appear during data fetching so that I know the system is working.	Create loading spinner component; Implement loading states; Add skeleton screens	Week 19	Programmer
T-079	EIS-066	As a User, I want success messages to appear as green notifications so that I know my	Design success notification styling; Implement notification system; Test user feedback	Week 20	Programmer

		actions completed successfully.			
T-080	EIS-067	As a User, I want error messages to appear as red notifications so that I'm aware of problems.	Design error notification styling; Implement error handling; Create error message templates	Week 20	Programmer
T-081	EIS-076	As a User, I want empty state screens to include helpful guidance and action buttons so that I know how to get started.	Design empty state components; Create empty state messages; Add call-to-action buttons	Week 20	Programmer
T-082	N/A	Project documentation - Search and UX	Document search functionality; Create filtering guide; Document loading state patterns	Week 19-20	Documentation Specialist
T-083	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review search and UX features	Week 19-20	Project Manager

SPRINT 11: DASHBOARD AND ANALYTICS FOUNDATION (Weeks 21-22)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-084	EIS-070	As a User, I want a dashboard that displays key metrics so that I can see important information at a glance.	Design dashboard layout; Create metric cards; Implement data aggregation	Week 21	Lead Programmer, Programmer
T-085	EIS-071	As a User, I want a calendar widget on the dashboard showing this month's activities so that I have quick access to schedules.	Design calendar widget; Implement mini calendar; Connect to calendar data	Week 21	Programmer
T-086	EIS-131	As a Legislative Member, I want to view the total number of scheduled sessions for the current month so that I can understand the workload.	Implement session count query; Create session metric display; Add time period filter	Week 21	Programmer
T-087	EIS-132	As a Legislative	Implement agenda count	Week 22	Programmer

		Member, I want to view the total number of pending agenda items so that I can prioritize my work.	query; Create agenda metric display; Add status filtering		
T-088	EIS-133	As a Legislative Member, I want to view the total number of overdue deadlines so that I can identify urgent tasks.	Implement overdue deadline query; Create deadline metric display; Add urgency indicators	Week 22	Programmer
T-089	EIS-135	As an Administrator, I want to view key performance indicators on the dashboard so that I can monitor overall performance.	Design KPI dashboard section; Implement KPI calculations; Create KPI visualizations	Week 22	Lead Programmer
T-090	N/A	Project documentation - Dashboard	Document dashboard metrics; Create dashboard user guide; Document KPI calculations	Week 21-22	Documentation Specialist

T-091	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review dashboard implementation	Week 21-22	Project Manager

SPRINT 12: SESSION AND AGENDA ANALYTICS (Weeks 23-24)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-092	EIS-136	As an Administrator, I want to view the total number of sessions conducted per month so that I can analyze session frequency trends.	Implement session analytics query; Create monthly session chart; Add trend visualization	Week 23	Lead Programmer
T-093	EIS-137	As an Administrator, I want to view session completion rates so that I can identify scheduling issues.	Implement completion rate calculation; Create completion rate display; Add comparison metrics	Week 23	Programmer

T-094	EIS-139	As an Administrator, I want to view sessions by type so that I can analyze session distribution.	Implement session type breakdown; Create distribution chart; Add filtering by type	Week 23	Programmer
T-095	EIS-140	As a Legislative Member, I want to view the total number of agenda items created per month so that I can track legislative activity.	Implement agenda analytics query; Create monthly agenda chart; Add activity timeline	Week 24	Programmer
T-096	EIS-141	As a Legislative Member, I want to view agenda item status distribution so that I can understand the workflow pipeline.	Implement status distribution query; Create status pie chart; Add workflow visualization	Week 24	Programmer
T-097	EIS-142	As an Administrator, I want to view average time from agenda item creation to approval so that I can	Implement time tracking analytics; Calculate average approval time; Create timeline visualization	Week 24	Lead Programmer

		identify bottlenecks.			
T-098	N/A	Project documentation - Analytics	Document analytics queries; Create analytics user guide; Document metric definitions	Week 23-24	Documentation Specialist
T-099	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review analytics implementation	Week 23-24	Project Manager

SPRINT 13: DEADLINE AND CALENDAR ANALYTICS (Weeks 25-26)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-100	EIS-145	As a Legislative Member, I want to view deadline compliance rate so that I can assess time management.	Implement compliance rate calculation; Create compliance chart; Add performance indicators	Week 25	Programmer
T-101	EIS-146	As an Administrator, I	Implement user-based deadline	Week 25	Lead Programmer

		want to view the total number of overdue deadlines by user so that I can identify individuals needing support.	query; Create user performance report; Add alert thresholds		
T-102	EIS-148	As a Legislative Member, I want to view upcoming deadlines for the next 7 days so that I can plan my work	Implement upcoming deadline query; Create deadline preview widget; Add sorting options	Week 25	Programmer
T-103	EIS-152	As an Administrator, I want to view calendar occupancy rate so that I can optimize scheduling.	Implement occupancy calculation; Create occupancy visualization; Add utilization metrics	Week 26	Programmer
T-104	EIS-154	As an Administrator, I want to view scheduling conflicts so that I can improve coordination.	Implement conflict detection algorithm; Create conflict report; Add resolution suggestions	Week 26	Lead Programmer
T-105	EIS-164	As a User, I want to view	Implement trend analysis; Create	Week 26	Programmer

		session trends in a line chart so that I can visualize activity patterns.	line chart component; Add date range filtering		
T-106	N/A	Project documentation - Deadline analytics	Document deadline metrics; Create compliance reporting guide; Document visualization types	Week 25-26	Documentation Specialist
T-107	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review analytics features	Week 25-26	Project Manager

SPRINT 14: VISUALIZATION AND REPORTING (Weeks 27-28)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-108	EIS-165	As a User, I want to view agenda status distribution in a pie chart so that I can quickly	Implement pie chart component; Create status distribution visualization; Add interactive tooltips	Week 27	Programmer

		understand the workflow state.			
T-109	EIS-166	As a User, I want to view deadline compliance in a bar chart so that I can track performance visually.	Implement bar chart component; Create compliance visualization; Add comparison periods	Week 27	Programmer
T-110	EIS-161	As an Administrator, I want to generate monthly summary reports in PDF format so that I can share performance data.	Implement PDF generation; Design report template; Add export functionality	Week 27	Lead Programmer
T-111	EIS-162	As an Administrator, I want to export analytics data to Excel/CSV format so that I can perform custom analysis.	Implement CSV export functionality; Add Excel export option; Create export templates	Week 28	Programmer
T-112	EIS-171	As a User, I want to filter all analytics by date range so that I can analyze	Implement date range filter component; Add filter persistence; Create preset date ranges	Week 28	Programmer

		specific periods.			
T-113	EIS-172	As an Administrator, I want to filter analytics by user or department so that I can analyze individual or team performance.	Implement user/department filters; Add multi-select functionality; Create filter combinations	Week 28	Programmer
T-114	N/A	Project documentation - Reporting	Document report generation; Create export guide; Document visualization library	Week 27-28	Documentation Specialist
T-115	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review reporting features	Week 27-28	Project Manager

SPRINT 15: AUDIT, LOGGING, AND DATA INTEGRITY (Weeks 29-30)

Task No.	User Story No.	User Stories	Tasks	Timeline	Responsible Team Member/s
T-116	EIS-005	As a System Administrator, I	Design audit log database	Week 29	Lead Programmer

		want to log all login attempts so that suspicious activities can be tracked.	schema; Implement login logging; Create log viewer interface		
T-117	EIS-009	As a System Administrator, I want to log all critical user actions with timestamp and user ID so that activities can be audited	Implement action logging middleware; Create audit trail system; Add timestamp tracking	Week 29	Lead Programmer
T-118	EIS-010	As a System Administrator, I want audit logs to be retained for at least 1 year so that historical investigations can be conducted.	Implement log retention policy; Create log archiving system; Add automated cleanup	Week 29	Lead Programmer
T-119	EIS-011	As a Compliance Officer, I want to generate audit reports for specific date ranges and users so that compliance reviews can be performed.	Design audit report interface; Implement report filtering; Add export functionality	Week 30	Programmer

T-120	EIS-013	As a Legislative Member, I want to receive notifications when agenda items or sessions I created are modified by others so that unauthorized changes are identified.	Implement change notification system; Create notification triggers; Add notification preferences	Week 30	Programmer
T-121	EIS-127	As a System Administrator, I want all data exchanges to be logged with timestamps and user identification so that integration activities can be audited.	Implement integration logging; Create integration audit trail; Add monitoring dashboard	Week 30	Lead Programmer
T-122	N/A	Project documentation - Audit system	Document audit logging; Create compliance guide; Document retention policies	Week 29-30	Documentation Specialist
T-123	N/A	Sprint planning and management	Conduct sprint planning; Daily standup coordination; Review audit implementation	Week 29-30	Project Manager

Table no. 9 Sprint Backlog (User Stories)

3.4.6.1 Sprint Burndown Chart

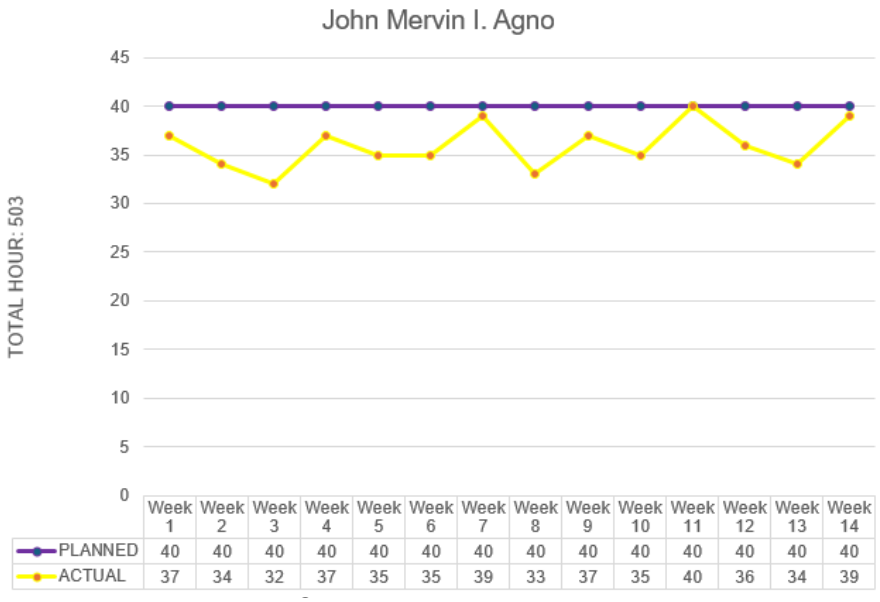


Figure no. 1 Burndown Chart

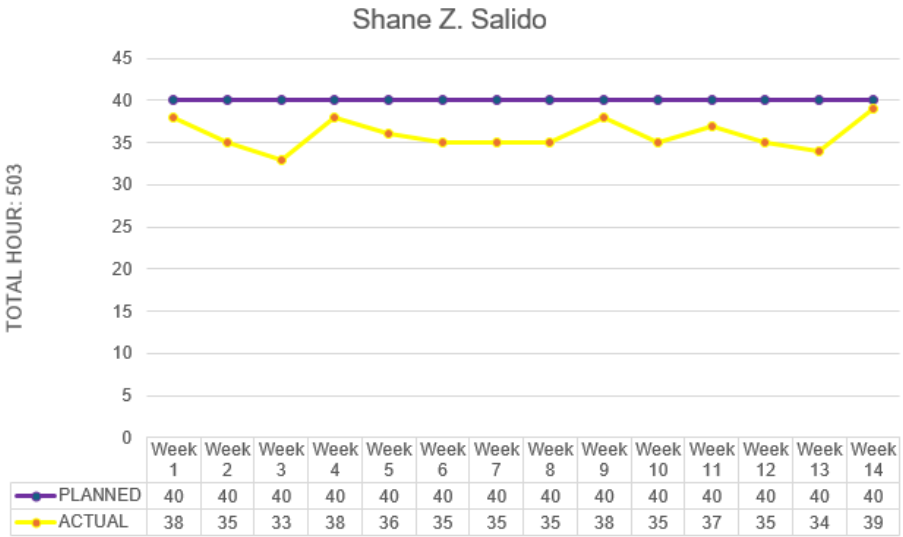
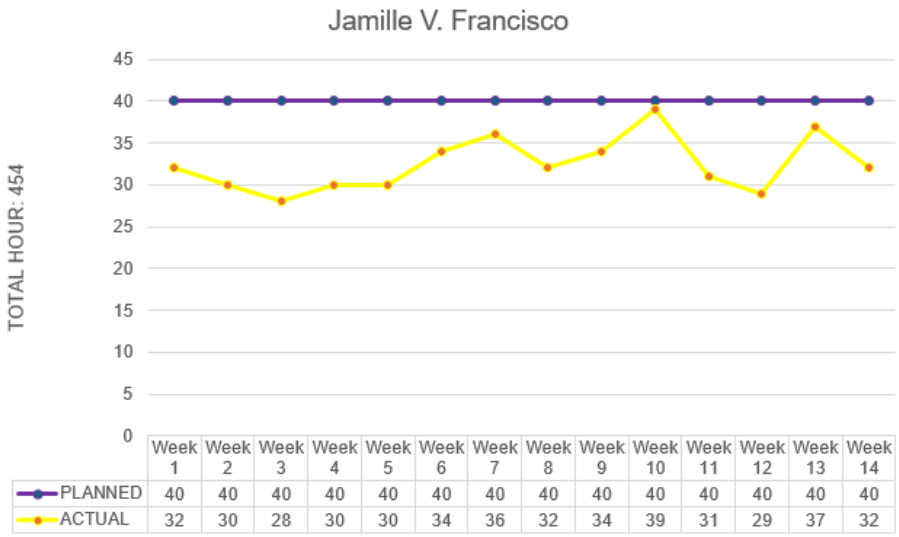
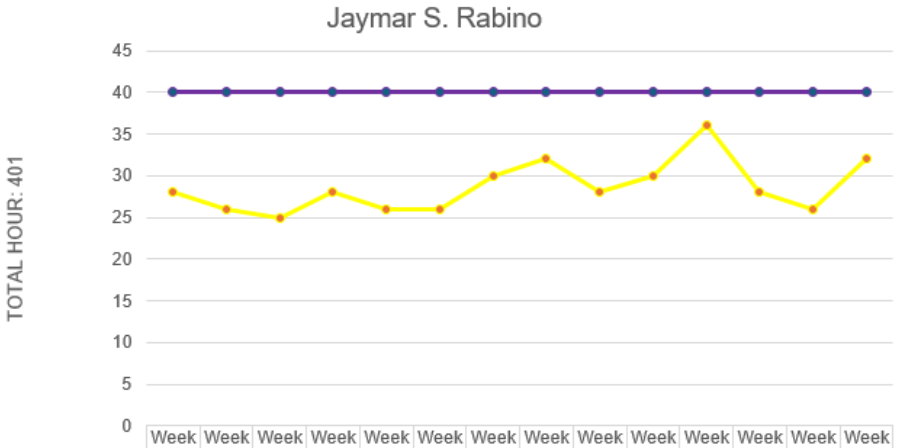


Figure no. 2 Burndown Chart



	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
PLANNED	40	40	40	40	40	40	40	40	40	40	40	40	40	40
ACTUAL	32	30	28	30	30	34	36	32	34	39	31	29	37	32

Figure no. 3 Burndown Chart



	Week 1	Week 2	Week 3	Week 4	Week 5	Week 6	Week 7	Week 8	Week 9	Week 10	Week 11	Week 12	Week 13	Week 14
PLANNED	40	40	40	40	40	40	40	40	40	40	40	40	40	40
ACTUAL	28	26	25	28	26	26	30	32	28	30	36	28	26	32

Figure no. 4 Burndown Chart

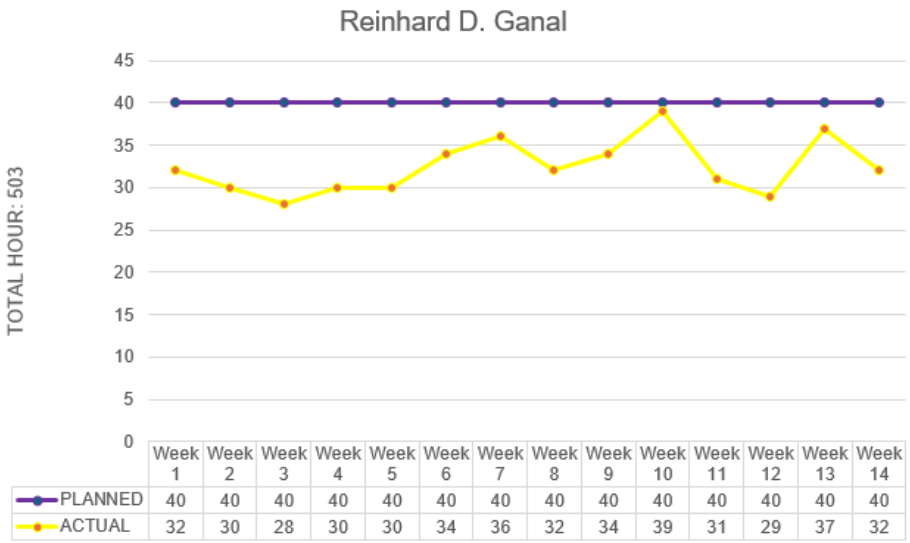


Figure no. 5 Burndown Chart

3.4.7. Increment

Sprint No.	Increment / Feature Delivered	User Story / Backlog Reference	Definition of Done (DoD) Criteria	Status	Remarks
Sprint 1	User Authentication System	EIS-001, EIS-002, EIS-003	✓ RBAC implemented with 4 user roles (Legislative Member, Committee Member, Administrative Staff,	Pending	Foundation for all system access

			Public) ✓ Password requirements enforced (min 8 chars, alphanumeric) ✓ Account lockout after 5 failed attempts functional ✓ Login page UI completed ✓ Unit tests passed (90% coverage) ✓ Security testing completed		
Sprint 1	Basic Calendar Display	CM-001, CM-002, CM-003	✓ Unified calendar view displays all events ✓ Daily, weekly, monthly views functional ✓ User-specific filtering works correctly ✓ Calendar UI responsive on desktop and mobile ✓ Integration tests passed ✓ User acceptance testing completed	Pending	Core module for system visibility
Sprint 1	Session Timeout and Login Logging	EIS-004, EIS-005	✓ Auto-logout after 30 minutes of inactivity ✓ Login attempts logged with timestamp and user ID	Pending	Essential security controls

			✓ Log storage functional ✓ Session management tested across browsers ✓ Security audit passed		
Sprint 2	Session Creation and Management	SS-001, SS-002, SS-003	✓ Session creation form functional ✓ Conflict detection algorithm working ✓ Presiding officer assignment implemented ✓ Session data saved to database ✓ Validation rules enforced ✓ Integration with Calendar Management complete ✓ User acceptance testing passed	Pending	Core Scheduling functionality
Sprint 2	Data Encryption Implementation	EIS-006, EIS-007	✓ HTTPS/TLS enabled on all pages ✓ Sensitive data encrypted in database ✓ Password hashing implemented (bcrypt/argon2) ✓ Encryption keys securely stored ✓ Security	Pending	Critical data protection

			penetration testing passed ✓ Performance impact assessed and acceptable		
Sprint 2	Automated Backup System	EIS-008, EIS-025	✓ Daily automated backups scheduled ✓ Backup stored in secure location ✓ Backup verification process functional ✓ Restore procedure documented ✓ Test restoration successful ✓ Monitoring alerts Configured	Pending	Data recovery Capability
Sprint 3	Agenda Creation and Proposal	AG-001, AG-002, AG-003	✓ Agenda item creation form functional ✓ Members can propose agenda items ✓ Agenda item fields validated ✓ Draft/Published status workflow working ✓ Agenda items linked to sessions ✓ Database relationships functional ✓ User acceptance	Pending	Core legislative function

			testing passed		
Sprint 3	Comprehensive Audit Logging	EIS-009, EIS-010, EIS-011	✓ All CRUD operations logged (agendas, sessions, deadlines) ✓ Logs include timestamp, user ID, action, affected data ✓ Log retention set to 1 year ✓ Audit report generation functional ✓ Log data protected from tampering ✓ Performance impact minimal ✓ Compliance review passed	Pending	Accountability and compliance
Sprint 3	Calendar Export Functionality	CM-005	✓ Export to PDF format functional ✓ Export to Excel format functional ✓ Exported data accurate and complete ✓ File download works on all browsers ✓ Format testing Passed	Pending	Pending
Sprint 4	Deadline Tracking System	DT-001, DT-002, DT-003	✓ Deadline creation and storage functional ✓ Automatic deadline calculation from sessions/agend as	Pending	Critical tracking functionality

			✓ Overdue deadline detection working ✓ Deadline status updates functional ✓ Integration with Calendar and Reminder modules complete ✓ Database triggers tested ✓ User acceptance testing passed		
Sprint 4	Input Validation and XSS Protection	EIS-012	✓ Input validation on all forms ✓ SQL injection prevention tested ✓ XSS attack prevention verified ✓ Parameterized queries implemented ✓ Security scanning passed (OWASP top 10) ✓ Error handling functional without exposing sensitive info	Pending	Essential security layer
Sprint 4	Change Notification System	EIS-013, DT-004	✓ Real-time notifications when items are modified	Pending	Transparency and awareness

			✓ Email notifications sent to creators ✓ In-app notification display functional ✓ Notification preferences saved ✓ Notification delivery tested ✓ User acceptance testing passed		
Sprint 5	Granular Access Permissions	EIS-015, EIS-016, EIS-017, EIS-018	✓ Calendar view permissions enforced ✓ Session creation restricted to authorized users ✓ Agenda approval workflow functional ✓ Voting records protected and immutable ✓ Permission checks in all endpoints ✓ Authorization testing passed ✓ Unauthorized access attempts blocked and logged	Pending	Fine-grained security control
Sprint 5	Automated Reminder System	RM-001, RM-002, RM-003, RM-004	✓ Reminder generation based on deadlines ✓ Email reminders sent 24 hours before	Pending	Proactive notification system

			events ✓ In-app reminders displayed ✓ Reminder scheduling algorithm functional ✓ User reminder preferences honored ✓ Delivery confirmation logged ✓ Integration testing passed		
Sprint 5	Agenda Version Control	EIS-014, AG-005	✓ Version history tracked for agenda items ✓ Version comparison view functional ✓ Rollback capability implemented ✓ Version metadata stored (date, user, reason) ✓ Version retrieval tested ✓ User acceptance testing passed	Pending	Change tracking and recovery
Sprint 6	Committee Management Integration	CM-004, SS-002, AG-006	✓ API connection to Committee Management established ✓ Committee schedules synced to	Pending	First external Integration

			calendar ✓ Conflict detection with committee meetings functional ✓ Committee recommendations pulled into agendas ✓ Data validation working ✓ Integration testing passed ✓ Error handling functional		
Sprint 6	Secure API Authentication	EIS-019, EIS-020	✓ API key authentication implemented ✓ API keys generated and stored securely ✓ Incoming data validation functional ✓ Invalid data rejection working ✓ API security testing passed ✓ Documentation completed	Pending	Secure integration foundation
Sprint 6	Calendar Search Functionality	CM-008	✓ Search by keyword functional ✓ Search by date range working ✓ Search by event type filtering accurate	Pending	Enhanced Usability

			✓ Search results displayed correctly ✓ Performance optimized for large datasets ✓ User acceptance testing passed		
Sprint 7	Secure File Upload System	EIS-022, EIS-023, EIS-024	✓ File type validation (PDF, DOCX, XLSX only) ✓ File size limit (25MB) enforced ✓ Virus scanning integrated ✓ Infected files rejected with user notification ✓ Files stored in secure location ✓ File upload testing passed ✓ Security audit completed	Pending	Document management security
Sprint 7	Public Hearing Integration	CM-004, SS-002, AG-007	✓ API connection to Public Hearing Management established ✓ Hearing schedules synced to calendar ✓ Public notice deadlines tracked ✓ Agenda items linked to hearings	Pending	Public engagement integration

			✓ Integration testing passed ✓ Data synchronization verified		
Sprint 7	Document Attachment to Agendas	AG-004	✓ File attachment to agenda items functional ✓ Multiple files can be attached ✓ File preview available ✓ File download working ✓ File metadata stored ✓ User acceptance testing passed	Pending	Enhanced agenda documentation
Sprint 8	Voting and Decision-Making Integration	CM-004, SS-007, AG-008	✓ API connection to Voting System established ✓ Voting schedules synced to calendar ✓ Agenda items linked to voting sessions ✓ Voting results updated in agenda status ✓ Vote immutability verified ✓ Integration testing passed ✓ Security audit completed	Pending	Decision tracking integration
Sprint 8	Calendar Event Color Coding	CM-007	✓ Color coding system implemented	Pending	Visual organization enhancement

			✓ Different colors for event types (sessions, hearings, deadlines, etc.) ✓ Color legend displayed ✓ User can customize colors ✓ Accessibility standards met (color blind friendly) ✓ User acceptance testing passed		
Sprint 8	Session Modification and Cancellation	SS-005, SS-006, SS-007	✓ Session modification form functional ✓ Change reason required and logged ✓ Session cancellation with notification working ✓ Affected users notified automatically ✓ Calendar updates reflected immediately ✓ Integration testing Passed	Pending	Schedule flexibility
Sprint 9	Legislative Archives Integration	CM-004, AG-009, SS-008	✓ API connection to Archives established ✓ Completed sessions archived	Pending	Historical record keeping

			automatically ✓ Finalized agendas sent to archives ✓ Archive retrieval functional ✓ Historical data accessible ✓ Integration testing passed ✓ Data integrity verified		
Sprint 9	Data Retention and Anonymization	EIS-028, EIS-030	✓ Retention policy configured (5 years) ✓ Automatic archiving after retention period ✓ Personal data anonymization in public reports ✓ Anonymization algorithm tested ✓ Compliance review passed ✓ Documentation completed	Pending	Privacy and Compliance
Sprint 9	Backup Restoration Testing	EIS-026, EIS-027	✓ Quarterly backup restoration procedure documented ✓ Test restoration successful ✓ Recovery time measured and acceptable ✓ Incident response	Pending	Disaster recovery readiness

			procedure documented ✓ Team training completed ✓ Runbook created		
Sprint 10	Public Consultation Integration	CM-004, AG-010, DT-005	✓ API connection to Consultation Management established ✓ Consultation schedules synced to calendar ✓ Feedback collection deadlines tracked ✓ Consultation results linked to agendas ✓ Integration testing passed ✓ User acceptance testing completed	Pending	Public participation Integration
Sprint 10	Advanced Reporting and Analytics	CM-009, SS-009, AG-011	✓ Session attendance reports generated ✓ Agenda completion metrics calculated ✓ Deadline adherence statistics displayed ✓ Custom date range selection working ✓ Export to PDF/Excel functional	Pending	Management insights

			✓ Performance optimized ✓ User acceptance testing passed		
Sprint 10	Meeting Management Integration	CM-004, SS-010	✓ API connection to Sessions and Meetings Management established ✓ Meeting schedules synced to calendar ✓ Attendance tracking integrated ✓ Meeting documentation linked ✓ Integration testing passed ✓ Data synchronization verified	Pending	Comprehensive meeting coordination
Sprint 11	Legislative Records Integration	CM-004, AG-012, DT-006	✓ API connection to Records and Research Analysis established ✓ Document deadlines synced ✓ Legislative trends pulled for agenda planning ✓ Analytics integration functional ✓ Research materials linked to agendas	Pending	Research and documentation integration

			✓ Integration testing passed ✓ Performance acceptable		
Sprint 11	API Integration Logging	EIS-021	✓ All API calls logged with timestamp and data ✓ Integration errors tracked ✓ Log analysis dashboard created ✓ Alert system for integration failures ✓ Log storage optimized ✓ Monitoring dashboard functional	Pending	Integration troubleshooting capability
Sprint 11	Privacy Request Management	EIS-029	✓ User data access request form functional ✓ Data retrieval and export working ✓ Request tracking system implemented ✓ Response within 30 days guaranteed ✓ User acceptance testing passed ✓ Legal compliance verified	Pending	Privacy rights compliance
Sprint 12	Ordinance Resolution Tracking Integration	CM-004, AG-013, SS-011	✓ API connection to Ordinance Tracking	Pending	Legislative process completion

			established ✓ Reading schedules synced to sessions ✓ Ordinance metadata pulled into agendas ✓ Document status updates reflected ✓ Voting schedules coordinated ✓ Integration testing passed ✓ End-to-end workflow tested		
Sprint 12	Enhanced Deadline Dashboard	DT-007, DT-008	✓ Visual dashboard showing all deadlines ✓ Priority flagging functional ✓ Overdue items highlighted ✓ Filtering and sorting working ✓ Real-time updates functional ✓ Mobile responsive ✓ User acceptance testing passed	Pending	Improved deadline visibility
Sprint 12	System-Wide Notification Preferences	RM-005, CM-010	✓ User notification preferences page functional	Pending	Personalized user experience

			✓ Email/SMS/In-app options configurable ✓ Notification frequency settings working ✓ Quiet hours respected ✓ Preferences saved and applied ✓ User acceptance testing passed		
Sprint 13	System Performance Optimization	Technical Debt	✓ Database query optimization completed ✓ Page load times < 3 seconds ✓ API response times < 1 second ✓ Caching implemented for frequent queries ✓ Load testing passed (100 concurrent users) ✓ Performance benchmarks documented	Pending	User Experience Enhancement
Sprint 13	Comprehensive Security Audit	EIS-060, EIS-061	✓ Code security review completed ✓ Vulnerability scanning performed ✓ Penetration testing conducted ✓ Security findings	Pending	Final security validation

			remediated ✓ Security documentation updated ✓ Third-party audit passed		
Sprint 13	User Acceptance Testing (UAT)	All User Stories	✓ UAT test cases executed (100% coverage) ✓ All critical bugs fixed ✓ User feedback incorporated ✓ Training materials prepared ✓ User manual completed ✓ Sign-off from stakeholders received	Pending	Production Readiness
Sprint 14	Production Deployment	System development	✓ Production environment configured ✓ SSL certificates installed ✓ Database migration successful ✓ Backup systems verified ✓ Monitoring tools deployed ✓ Rollback plan tested ✓ Deployment checklist completed	Pending	System go-live
Sprint 14	User Training and Documentation	Training and Support	✓ Administrator training conducted	Pending	User enablement

			✓ End-user training sessions completed ✓ Video tutorials created ✓ Help documentation published ✓ Support desk procedures established ✓ FAQ documentation created		
Sprint 14	Post-Launch Support and Monitoring	Maintenance	✓ 24/7 monitoring active ✓ Support ticketing system operational ✓ Incident response team assigned ✓ Performance metrics dashboard live ✓ Maintenance Schedule established	Pending	Ongoing Operations

Table no. 4 Increment

3.5 Microservices Architecture

This microservices architecture is designed to ensure modularity, scalability, and operational resilience. This architectural approach allows each service to operate, scale, and be updated independently without affecting the overall system. By isolating functions, the system minimizes the risk of complete service disruption and supports future enhancements, including integration with other LGU systems such as the Blockchain-Enabled Public Hearing Management System. The microservices model is particularly suitable for legislative environments where processes evolve over time and require flexible system adaptation.

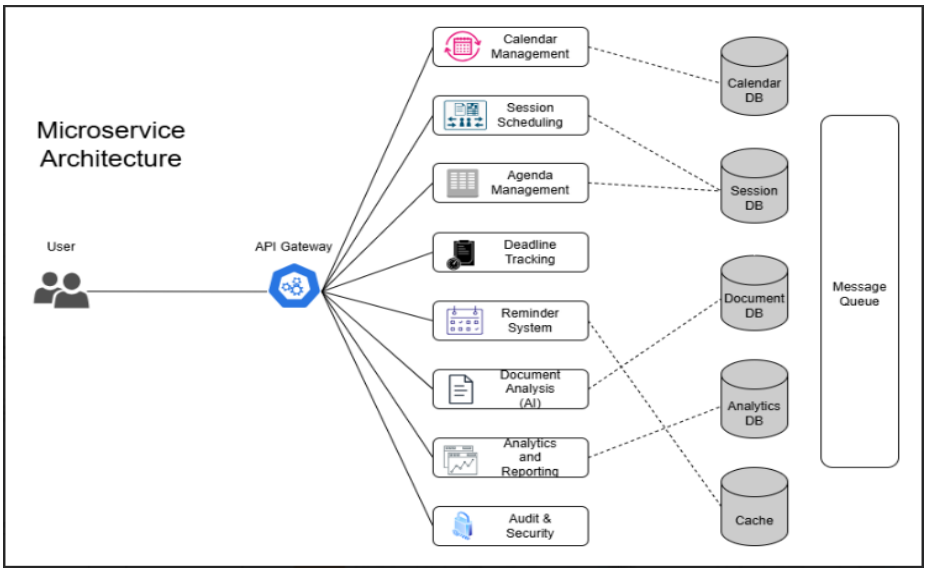


Figure no. 6 Microservice Architecture

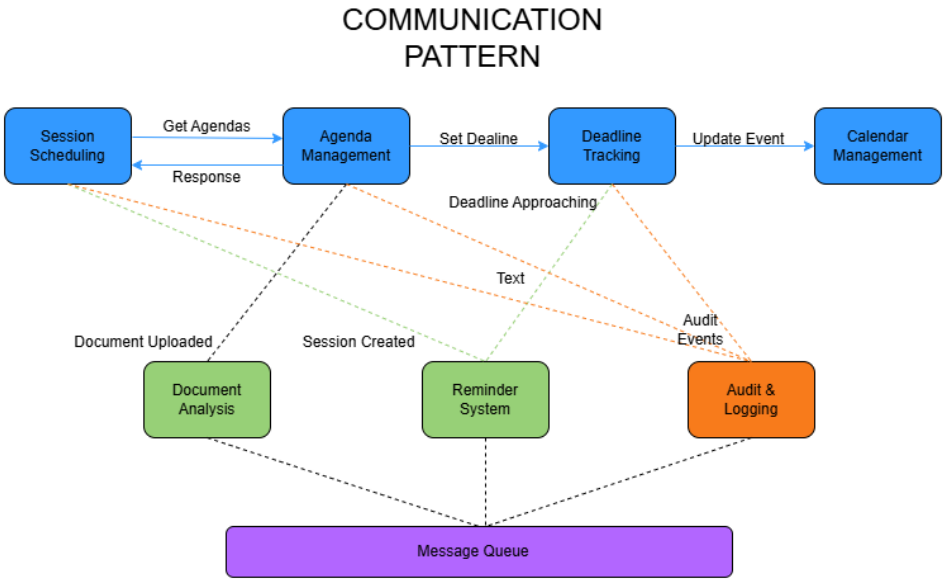


Figure no. 7 Communication Pattern for Microservices

Here's the **communication pattern diagram** for microservices:

- 🔵 **Blue solid lines → Synchronous (Request/Response)** calls.
- 🟢 **Green dashed lines → Asynchronous (Event-driven)** interactions.

The communication pattern of the Legislative Agenda and Calendar Management System illustrates how different microservices interact using a combination of synchronous and asynchronous messaging to support efficient legislative operations. The process begins with the Session Scheduling service requesting agenda data from the

Agenda Management service through synchronous communication, ensuring real-time retrieval and response of agenda information. Once an agenda is confirmed, the Agenda Management service forwards scheduling details to the Deadline Tracking service, which is responsible for setting and monitoring legislative deadlines. As deadlines are defined or updated, the Deadline Tracking service synchronously communicates with the Calendar Management service to update legislative events and ensure that all schedules remain consistent across the system.

Level 1 DFD (Context Diagram)

This shows the **entire legislative agenda and calendar management system as one process** and how external entities interact with it.

- **External Entity:** Calendar Management, Session Scheduling,
- **System (Process):** Legislative Agenda and Calendar Management System, Agenda Creation and Management, Deadline Tracking and Monitoring, Reminder System
- **Data Stores:** Calendar DB, Session DB, Agenda DB, Deadline DB, Notification DB
- **Data Flows:** Calendar Setup, Calendar Records, Calendar Data, Session Schedules,

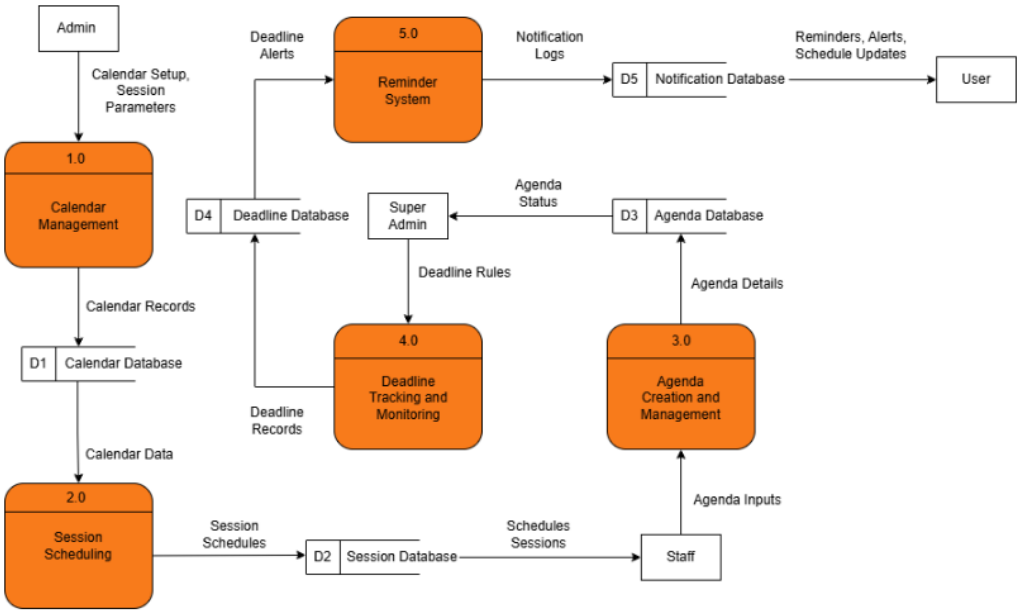


Figure no. 8 Data Flow Diagram for Microservices

3.6 DevOps Implementation

To ensure rapid delivery of improvements, pipeline automates the process of building, testing, and deploying system updates. When developers commit changes to the source code repository, automated build and test processes are triggered to verify system stability and prevent integration errors. Once validated, the system is automatically deployed to the cloud environment without disrupting ongoing legislative operations.

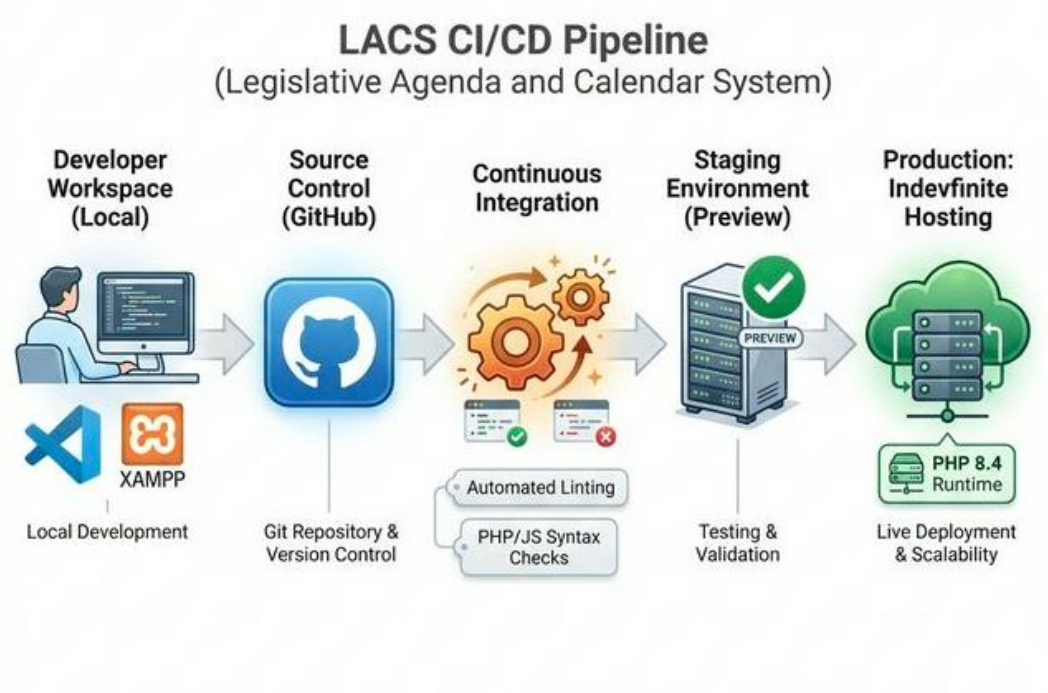


Figure no. 9 CI/CD Pipeline

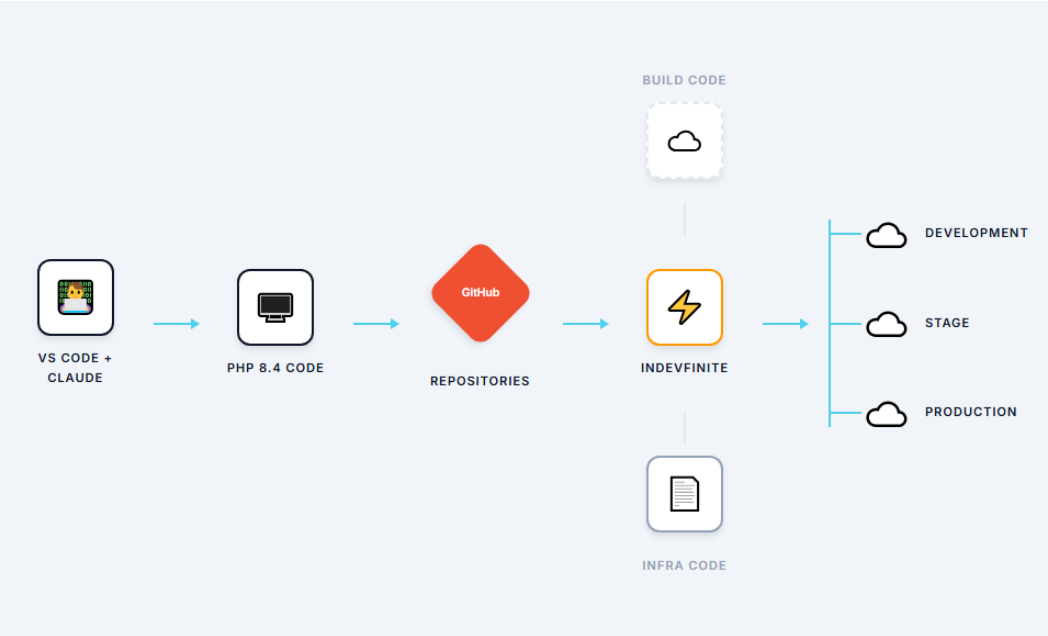


Figure no. 10 Infrastructure as a Code (IaC)

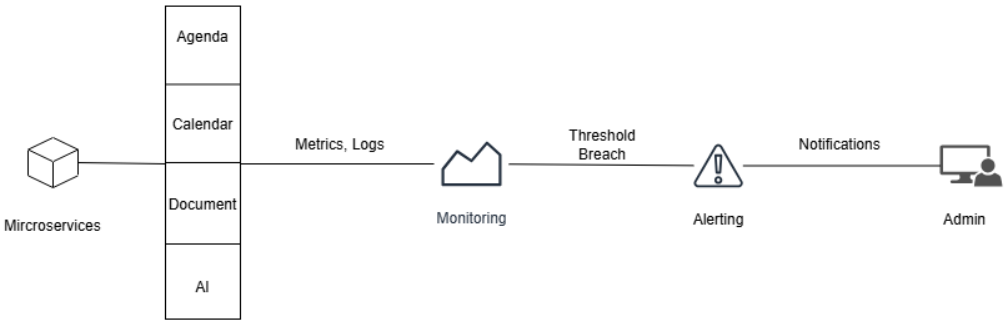


Figure no. 11 Monitoring and Alerting

We made this Monitoring and Alerting architecture to ensure the reliability and continuous operation of the Legislative Agenda and Calendar Management System. All microservices generate system metrics and logs that are collected by the monitoring system to track performance, availability, and errors in real time. When predefined thresholds such as service downtime or high error rates are exceeded, the monitoring system triggers the alerting service. Alerts are then sent to system administrators, enabling timely response and minimizing disruptions to legislative operations. This architecture supports proactive system management and helps maintain stable and uninterrupted system services.

3.7 Integration Approach for Information Systems

The Legislative Agenda and Calendar Management System adopts a comprehensive Business Process Architecture (BPA) integration approach to unify legislative planning, scheduling, documentation, and monitoring into a single, interconnected information ecosystem. Rather than functioning as isolated modules, the system is designed so that each legislative function is tightly integrated, allowing information generated in one process to seamlessly flow into subsequent processes. This approach ensures continuity, accuracy, and efficiency across all stages of legislative operations.

The diagram illustrates a lifecycle-based integration strategy that links pre-legislative activities, core legislative operations, and post-legislative tracking into one coherent workflow. Pre-legislative processes such as committee profiling, legislative trend analysis, document storage and versioning, and advanced document analysis directly feed into agenda creation and session scheduling. Once an agenda is finalized, it is automatically synchronized with calendar management, deadline tracking, and meeting coordination modules, ensuring that all schedules, deadlines, and responsibilities are consistently aligned. The integration further extends to deliberation, voting sessions, action item follow-ups, and document finalization,

where outputs are systematically recorded, tracked, and archived. This seamless integration eliminates data silos, enhances traceability of legislative actions, and ensures that all agenda and calendar records remain accurate, timely, and transparent throughout the legislative lifecycle.

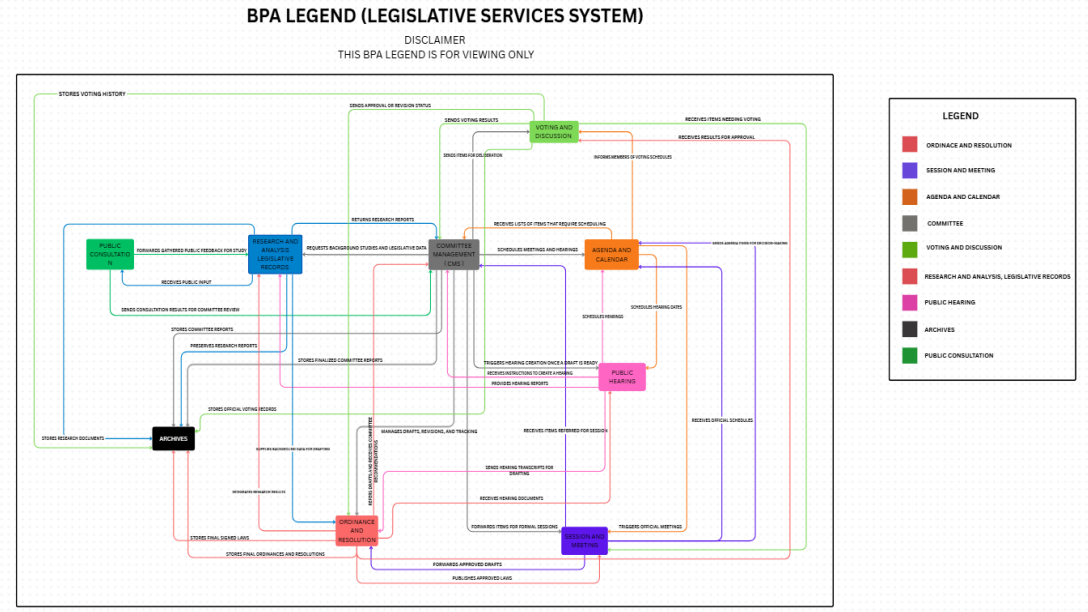


Figure no. 12 Business Process Architecture Level 0

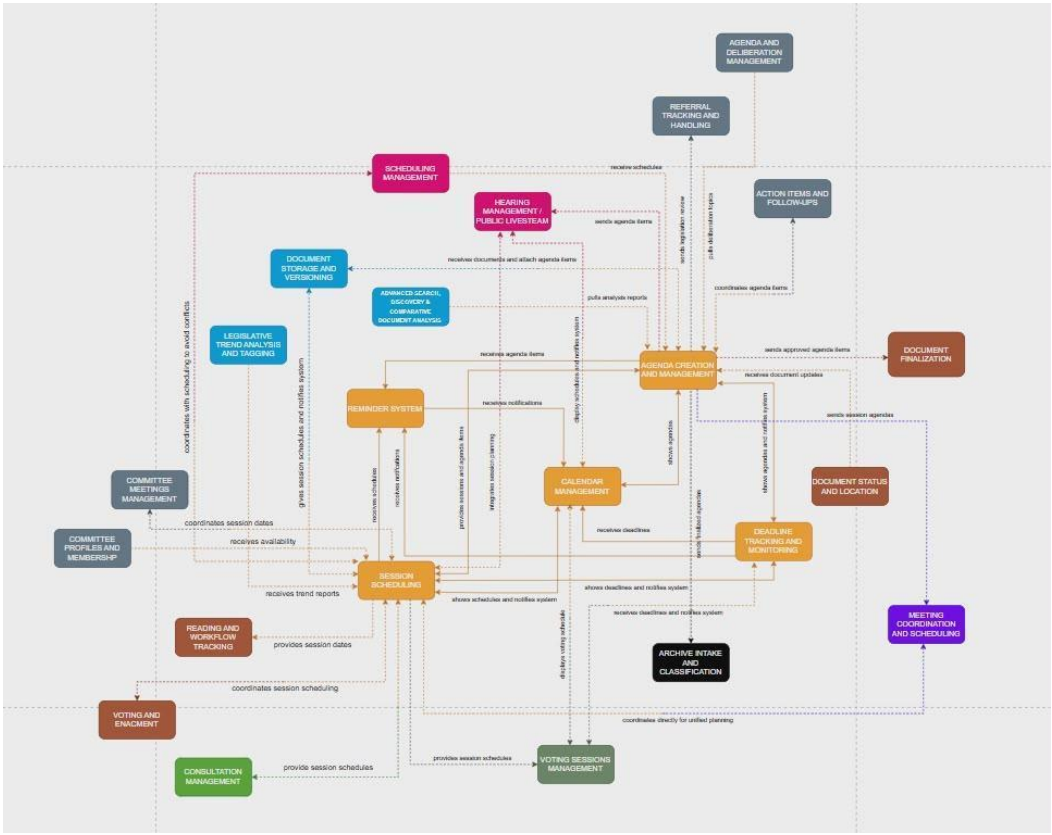


Figure no. 13 Business Process Architecture Level 1

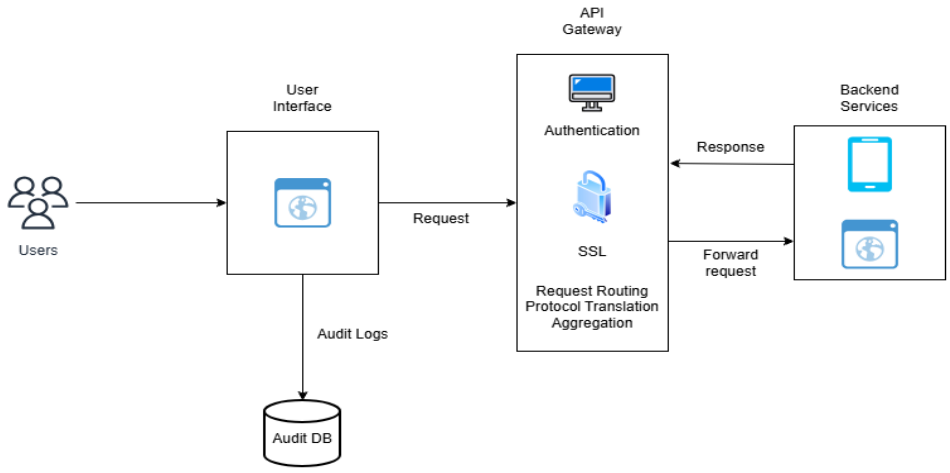


Figure no. 14 API Gateway

This API Gateway architecture serves as a centralized entry point for all client requests to the Legislative Agenda and Calendar Management System. Users access the system through mobile or web applications, and all requests are routed through the API Gateway before reaching backend services. The gateway is responsible for authentication, SSL security, request routing, protocol translation, and request aggregation, ensuring secure and controlled access to system resources. Additionally, the API Gateway records audit logs of system transactions to support accountability and traceability.

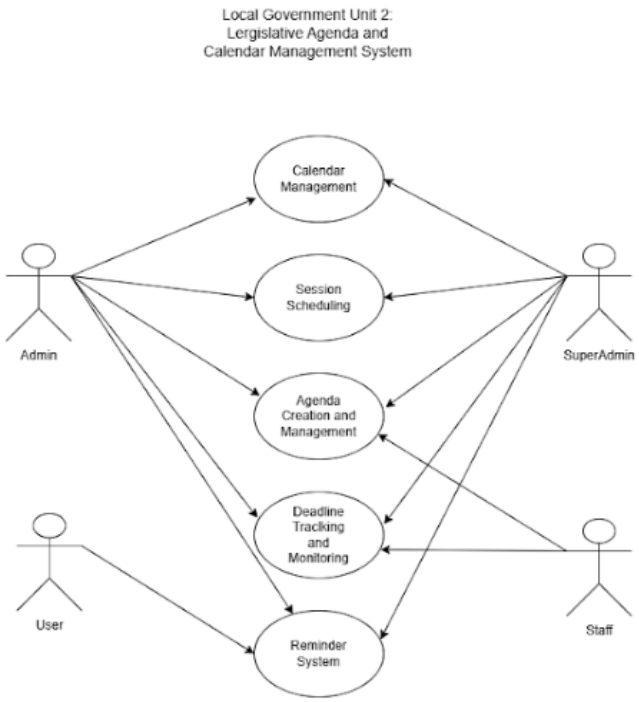


Figure no. 12 Use Case Diagram

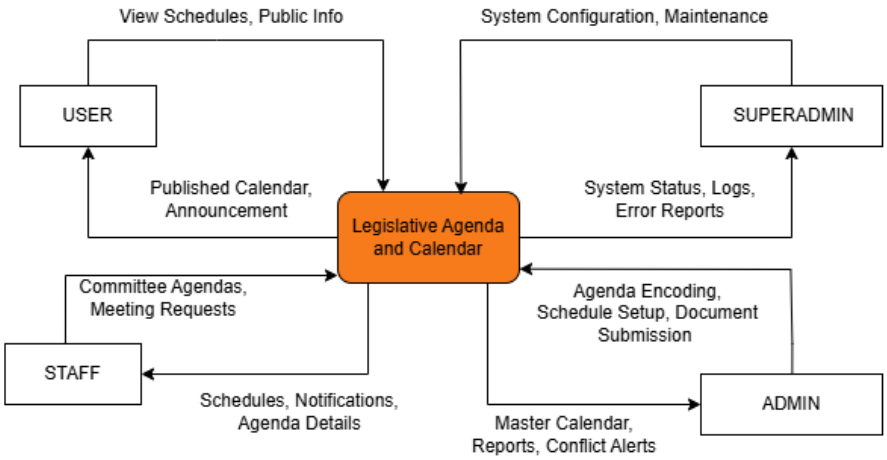


Figure no. 15 Data Flow Diagram

The context diagram illustrates the overall interaction between external entities and the Legislative Agenda and Calendar System. Users are able to view published schedules and public information, while staff members submit committee agendas and meeting requests and receive schedules, notifications, and agenda details from the system. Administrators are responsible for encoding agendas, managing schedules, and submitting documents, and in return they receive the master calendar, reports, and conflict alerts.

Super administrators oversee system configuration and maintenance and monitor system status, logs, and error reports. This diagram defines the system boundary and emphasizes how different user roles interact with the system at a high level without showing internal processes.

The Level 1 Data Flow Diagram provides a more detailed view of the internal operations of the system by decomposing it into major processes and data stores. Calendar Management handles calendar setup and parameters and stores calendar records in the calendar database, which are then used by the Session Scheduling process to generate session schedules stored in the session database. Agenda Creation and Management processes agenda inputs from staff and saves agenda details in the agenda database. Deadline Tracking and Monitoring applies deadline rules, tracks submissions, and updates deadline records in the deadline database. The Reminder System generates deadline alerts and notifications and stores notification logs in the notification database. Together, these processes show how data flows within the system to support agenda management, scheduling, deadline monitoring, and automated notifications.

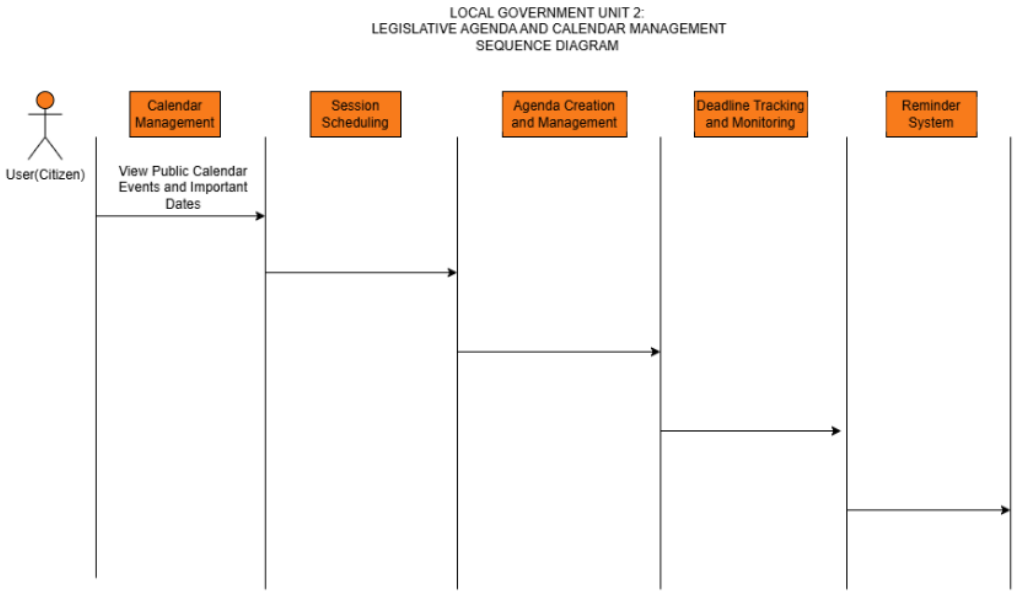


Figure no. 16 Sequence Diagram

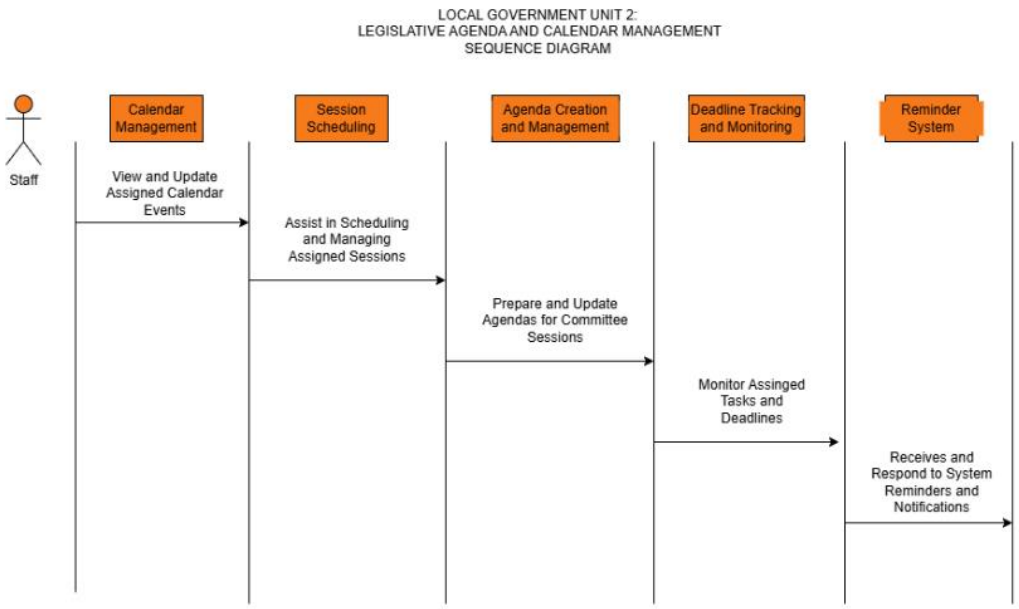


Figure no. 17 Sequence Diagram

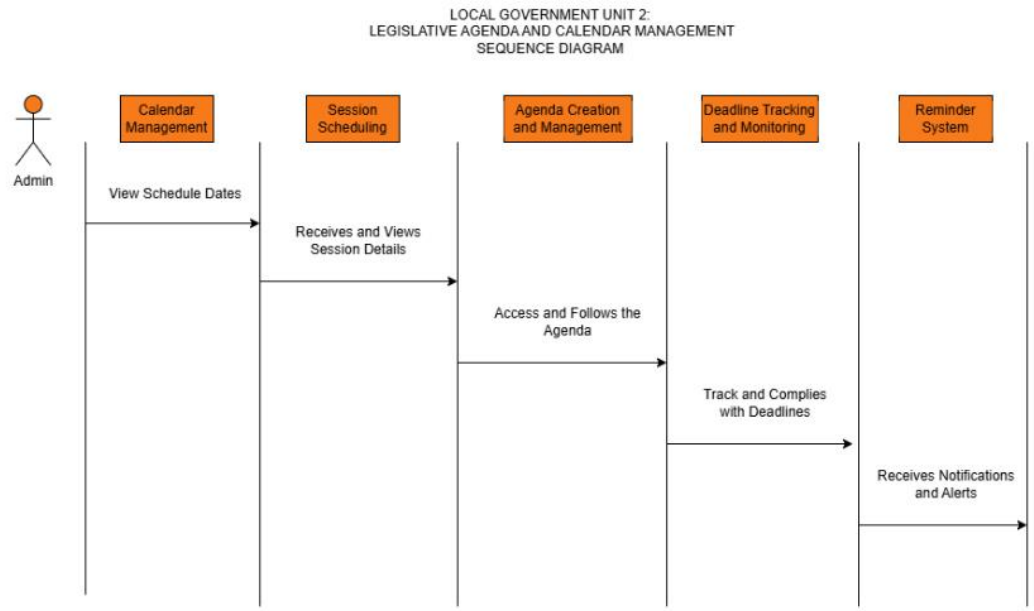


Figure no. 18 Sequence Diagram

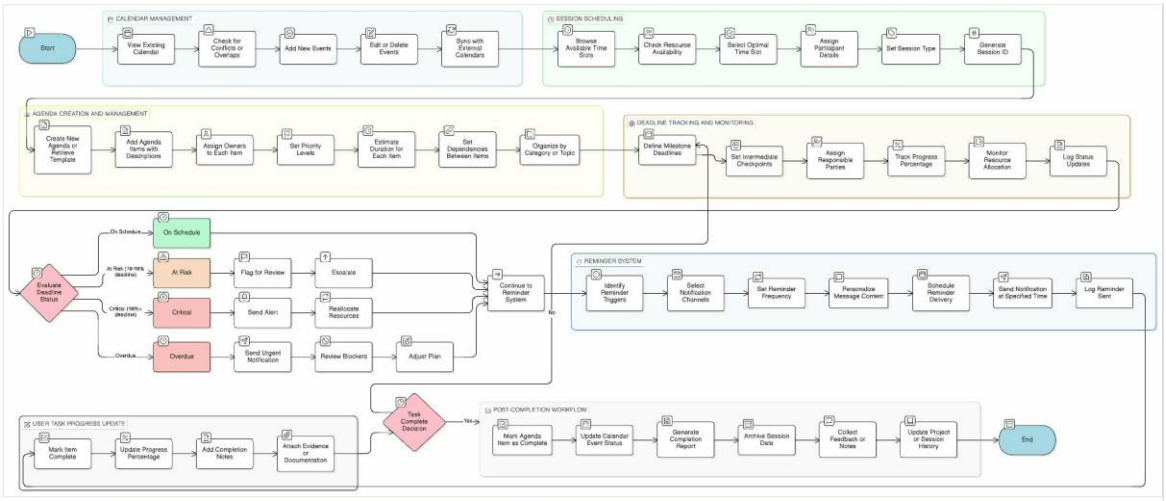


Figure no. 19 Flowchart

This flowchart illustrates the overall workflow of the Legislative Agenda and Calendar System, beginning with calendar management where users view existing calendars, check for conflicts, and add, edit, or synchronize events. Session scheduling then identifies available time slots, checks resource availability, selects optimal schedules, assigns participants, and generates session identifiers. Agenda creation and management allows users to define agenda items, assign owners, set priorities, estimate durations, and establish dependencies. Deadline tracking and monitoring evaluates task status by defining milestones, assigning responsibilities, tracking progress, and categorizing tasks as on schedule, at risk, critical, or overdue, which determines the appropriate actions and escalation. The reminder system generates and sends notifications based on deadline status, while users update task progress and submit completion details. Once tasks are completed, post-completion processes update calendar statuses, generate reports, archive data, and record feedback, completing the workflow.

3.8 Introduction to TOGAF and the Four Architectural Domains

1. Business Architecture

The Business Architecture defines how the Legislative Agenda and Calendar Management System supports the daily operations and governance of the legislative office. As reflected in the business process workflows, this domain outlines core activities such as Agenda Creation, Session Scheduling, Deadline Tracking, and Post-Session Reporting. It identifies key actors including Admin, Staff, Superadmin, and Users, and clarifies how each role interacts with the system to ensure timely coordination of legislative sessions, transparency of schedules, and accountability in task execution. By aligning system functions with real legislative procedures, the Business Architecture ensures that the platform directly supports efficient decision-making and organized legislative operations.

2. Data Architecture

The Data Architecture governs how legislative information is structured, stored, and maintained throughout

the system. This domain is represented through data flow diagrams and logical data stores that manage critical data such as calendar events, agenda items, session details, deadlines, notifications, and progress records. It ensures that data flows seamlessly between modules like Agenda Management, Session Scheduling, and Deadline Monitoring while maintaining consistency and integrity. Proper data organization enables accurate tracking of agenda status, historical session records, and reliable notification delivery, supporting both operational efficiency and long-term record keeping.

3. Application Architecture

The Application Architecture defines the internal structure and interaction of the software components that make up the Legislative Agenda and Calendar Management System. The system is organized into distinct but interconnected application domains such as Calendar Management, Agenda Creation and Management, Session Scheduling, Deadline Tracking and Monitoring, Reminder System, and Post-Completion Workflow. Each domain encapsulates its own business logic and processes, allowing

the system to reflect real-world legislative workflows. This modular structure ensures that enhancements or changes—such as improving reminder automation or deadline evaluation logic—can be implemented without disrupting other system functions, resulting in a scalable and maintainable application.

4. Technology Architecture

The Technology Architecture outlines the technical infrastructure required to deploy and operate the system reliably and securely. This domain includes the use of web-based application technologies, centralized databases, notification services, and server infrastructure that support real-time scheduling, monitoring, and alerts. The architecture ensures system availability, data security, and performance optimization, particularly for deadline tracking and notification delivery. By leveraging modern deployment practices and scalable infrastructure, the Technology Architecture guarantees that the system can support continuous legislative operations while adapting to future growth and technological improvements.

Chapter 4
REQUIREMENTS AND ANALYSIS

4.1 Stakeholder Identification

The successful design and development of the Local Government Unit 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-Classification required the identification of all individuals and groups who have an interest in, or are affected by, the system. Stakeholders were identified through direct engagement with the Valenzuela City Sangguniang Panlungsod, particularly through the interview conducted with Mr. Jojo Faustino, and through analysis of the legislative workflow described in Chapter 1. These stakeholders were categorized into internal stakeholders, who interact with the system directly in the course of legislative operations, external stakeholders, who are affected by or benefit from the system's outputs, and project stakeholders, who are directly involved in the system's development and evaluation.

Table 11 presents the identified stakeholders, their classification, and their respective stake or role in the system.

Stakeholder	Category	Interest/Stake	Role in the System
Stakeholder Administrator	Internal	Manages system configuration, user accounts, and access permissions	Full administrative access; configures modules and monitors system health
Superadmin	Internal	Oversees overall system governance and top-level configuration	Highest level of access; manages Admin accounts and system-wide settings
Staff (Sangguniang Panlungsod Secretary's Office)	Internal	Encodes agendas, submits documents, manage day-to-day scheduling	Primary day-to-day user; creates and manages agendas, sessions, and documents
Legislative/ Committee Members	Internal	Requires timely, accurate information on sessions, deadlines, and assigned tasks	Views schedules, receives notifications, participates in deliberation and voting workflows
Mr. Jojo Faustino (Administrative Staff, Product Owner representative)	Internal/ Project	Represents the operational needs of the Sangguniang Panlungsod Office	Primary requirements source; validates features against real office workflow

Valenzuela City Sangguniang Panlungsod (institution)	External	Institutional efficiency, compliance with legislative deadlines, transparency	Benefiting organization; adopts the system for legislative operations
Valenzuela City Constituents/Public Users	External	Access to published legislative schedules and ordinance status	Views public facing schedules and ordinance progress through the transparency feature
Other Philippine LGUs	External	Potential adopters seeking a replicable model for legislative digitalization	Indirect beneficiaries through documentation and architecture as a reference model
Research Adviser (Mr. Ronald Roldan Jr.)	Project	Ensures academic and technical soundness of the study	Provides guidance, reviews methodology and documentation
Development Team (Project Manager, Lead Programmer, Programmer, Documentation Specialists)	Project	Responsible for designing, building, testing, and documenting the system	Executes the Scrum roles defined in Table 2
Panelists	Project	Evaluate the technical and academic merit	Review and approve the system during of the capstone oral defense

4.2 Requirements Gathering Techniques

The successful Requirements for the Local Government Unit 2 Legislative Agenda and Calendar Management System were gathered using a combination of qualitative and iterative techniques suited to the Agile Scrum methodology adopted for the project, as introduced in Chapter 3.

Key Informant Interview. The primary requirements-gathering technique used was a structured interview with Mr. Jojo Faustino, Administrative Staff of the Sangguniang Panlungsod Office, who represented the actual end-users of the system. This interview, referenced in Chapter 1.1, surfaced the core operational problems, the absence of a unified digital system, the difficulty of managing committee workloads, inefficient meeting and session scheduling, and delays in the manual processing of legislative documents, that directly shaped the Problem Statement (Table 1) and the Product Backlog.

Document and Process Review. The research team reviewed the existing manual and paper-based processes used by the Sangguniang Panlungsod for agenda creation, scheduling, and document handling. This review clarified the legislative reading stages (First Reading, Second

Reading, Third Reading) referenced in Objective 3 and informed how these stages needed to be represented within the system's workflow.

Iterative Backlog Refinement. Consistent with the Agile Scrum methodology described in Chapter 3, requirements were not gathered only once at the start of the project but were continuously refined across sprints. As the development team implemented early features and reviewed them with the Product Owner representative, new user stories and refinements were added to the Product Backlog (Tables 6 to 9), allowing requirements to evolve based on feedback from working increments of the system rather than a single static requirements document.

Sprint Reviews. At the close of each sprint, completed features were reviewed against stakeholder expectations, which served as an additional validation point for previously gathered requirements and a source of newly discovered requirements, particularly around notification behavior, access permissions, and document classification accuracy.

These combined techniques ensured that the requirements captured in this chapter reflect both the stated needs of the Sangguniang Panlungsod (through direct interview) and the practical, evolving needs uncovered during actual development (through iterative Scrum practices).

4.3 User Stories and User Cases

4.3.1 User Stories

Table 12: Representative User Stories by Module

Module	User Story	Priority
Calendar Management	As a Staff member, I want to view and manage the legislative calendar, so that I can schedule sessions without conflicts.	High
Session Scheduling	As a Staff member, I want the system to detect scheduling conflicts automatically, so that overlapping sessions are prevented.	High
Agenda Creation and Management	As a Staff member, I want to create and edit agenda items with assigned owners and priorities, so that responsibilities are clear.	High
Deadline Tracking and Monitoring	As a System Administrator, I want the system to track legal deadlines automatically, so that reading and publication schedules are not missed.	High
Document Analysis/ AI Classification	As a System Administrator, I want the system to track legal deadlines automatically, so that reading and publication schedules are not missed.	High
Notification/Reminder	As a Committee Member, I want to receive automated reminders about upcoming deadlines and sessions, so that I do not miss required actions.	Meduim

Access Control	As a System Administrator, I want to manage granular access permissions, so that users only access information relevant to their role.	Medium
Transparency/Public Access	As a Constituent, I want to view published legislative schedules and ordinance status, so that I can stay informed about local governance.	Meduim
Audit Logging	As a Superadmin, I want the system to maintain comprehensive audit logs, so that all actions can be traced for accountability.	High

4.3.1 User Stories

Table 13: Use Case Description to Create Agenda

Field	Description
Use Case ID	UC-01
Actor	System (automated), Admin
Description	Automatically monitors legislative deadlines and flags approaching or overdue items.
Preconditions	Staff is authenticated and has agenda-creation permissions.
Main Flow	1. System evaluates deadline rules against current date. 2. System categorizes item status (on schedule, at risk, critical, overdue). 3. System triggers Reminder Service if thresholds are met.
Postconditions	Agenda item is stored and made available for scheduling.

Table 14: Use Case Description to Track Deadline

Field	Description
Use Case ID	UC-02
Actor	Staff
Description	Allows Staff to create and submit a new legislative agenda item.
Preconditions	Staff is authenticated and has agenda-creation permissions.
Main Flow	1. System evaluates deadline rules against current date. 2. System categorizes item status (on schedule, at risk, critical, overdue). 3. System triggers Reminder Service if thresholds are met.
Postconditions	Deadline status is updated and, if applicable, a notification is dispatched.

Table 15: Use Case Description to Classify and Summarize Document

Field	Description
Use Case ID	UC-03
Actor	Staff, System (AI-Classification module)
Description	Analyzes an uploaded legislative document, classifies it by type, and generates a summary.
Preconditions	Document has been uploaded to the system.
Main Flow	1. Staff uploads a document. 2. Document Analysis module preprocesses the document text. 3. AIclassification model categorizes the document (e.g., ordinance, resolution, committee report, minutes). 4. System generates a summary of key content. 5. Classification and summary are attached to the relevant agenda item.
Postconditions	Document is classified, summarized, and linked to its corresponding agenda record.

Table 16: Use Case Description to Receive Notification

Field	Description
Use Case ID	UC-04
Actor	Staff, Committee Member
Description	Delivers automated reminders regarding upcoming sessions or approaching deadlines.
Preconditions	User has an active account with notification preferences configured.
Main Flow	1. Reminder Service detects a triggering condition (e.g., deadline approaching). 2. System generates a notification. 3. Notification is delivered to the relevant user(s). 4. Notification event is logged.
Postconditions	User is informed of the pending action; notification log is updated.

Table 17: Use Case Description to View Public Schedule

Field	Description
Use Case ID	UC-04
Actor	User (Constituent)
Description	Allows the public to view published legislative schedules and ordinance status.
Preconditions	Schedule/ordinance has been marked as public by Staff or Admin.
Main Flow	1. Constituent accesses the public portal. 2. System retrieves published schedule and ordinance data. 3. Constituent views the information
Postconditions	Constituent has current, accurate information without requiring internal system access.

4.4 Functional Requirements for Integration

This section defines the functional requirements governing how the system's internal modules and external systems must interoperate, distinct from the feature-level user stories in Section 4.3. These requirements build directly on the Integration Approach for Information Systems introduced in Chapter 3.7 and the API Gateway architecture presented in Figure 14.

Internal Module Integration Requirements

- FR -INT-01: The system shall synchronize a finalized agenda with the Calendar Management module in real time, ensuring that session dates and details are immediately reflected on the master calendar.

- FR -INT-02: The system shall propagate deadline data generated during Agenda Creation to the Deadline Tracking and Monitoring module without requiring manual re-entry.

- FR -INT-03: The system shall trigger the Reminder System automatically whenever a monitored deadline crosses a predefined threshold (e.g., approaching, at risk, overdue), consistent with the flowchart described in Chapter 3.7.

- FR -INT-04: The system shall attach the output of the Document Analysis/AI-Classification module (classification label and summary) to its corresponding agenda item automatically upon completion of processing.

- FR -INT-05: The system shall record all inter-module transactions (agenda updates, deadline changes, notification dispatches) in a centralized audit log accessible to the Superadmin.

External Module Integration Requirements

The Legislative Agenda and Calendar System (LACS) does not operate in isolation — it is one of seven interconnected systems that together make up the LGU 2 legislative information ecosystem: the Ordinance and Resolution Tracking System (ORTS), LACS, the Committee Management System (CMS), the Public Hearing Management System (PHMS), the Public Consultation Management System (PCMS), the Legislative Records Management System (LRMS), and the Legislative Archiving System (LAS). Within this ecosystem, LACS functions as the scheduling and calendar authority: it receives triggers from OR TS and CMS and, in turn, feeds confirmed dates back out to CMS and PHMS.

- FR -INT-06 (from ORTS): The system shall receive new ordinance/resolution proposal data from OR TS and automatically generate

a corresponding agenda-slot request within the Agenda Creation and Management module.

- FR -INT-07 (from ORTS): The system shall receive reading-stage data (First Reading, Second Reading, Third Reading) from ORTS and schedule each reading directly on the Calendar Management module, without requiring manual re-entry by Staff.

- FR -INT-08 (to ORTS): Once an agenda placement locks a session date for a given ordinance, the system shall notify OR TS so that the ordinance's status and location can be updated accordingly.

- FR -INT-09 (from CMS): The system shall receive committee routing data from CMS whenever an ordinance is assigned to a committee, and shall automatically generate the corresponding review deadline within the Deadline Tracking and Monitoring module.

- FR -INT-10 (to CMS): The system shall dispatch deadline reminders generated by the Reminder System to CMS for delivery to the relevant committee members.

- FR -INT-11 (from CMS): The system shall accept emergency meeting requests submitted by CMS and write them directly to the Calendar Management module.

- FR -INT-12 (to PHMS): The system shall push confirmed public hearing session dates and times to PHMS immediately upon confirmation, to support PHMS 's automated generation of public notices.

- FR -INT-13: The system shall support calendar export in standard formats (e.g., iCalendar) to allow synchronization with external calendar applications used by committee members, consistent with the "Calendar Export Functionality" identified in the Sprint Board (Table/Board, Chapter 3.3).

- FR -INT-14: All requests to or from ORTS, CMS, and PHMS shall pass through the API Gateway for authentication, SSL enforcement, and audit logging before reaching any backend LACS microservice, consistent with the API Gateway architecture in Figure 14.

LACS do not integrate directly with PCMS, LRMS, or LAS — these sit further downstream in the ecosystem (PCMS handles public consultation queuing off PHMS; LRMS and LAS handle document archival originating from ORTS, CMS, and PHMS) and fall outside LACS's own integration boundary, though they are documented in Appendix A.2 for architectural completeness.

External Module Integration Requirements

- FR -INT-09: The system shall maintain referential consistency across the Calendar, Session, Agenda, Deadline, and Notification databases such that a change in one data store (e.g., a rescheduled session) is reflected in all dependent modules within an acceptable synchronization window.

- FR -INT-10: The system shall handle integration failures (e.g., a downstream service unavailable) gracefully, queuing affected events for retry rather than silently discarding them, to preserve legislative data integrity.

BUSINESS PROCESS ARCHITECTURE

5.1 Identification of Business Process

Legislative Agenda and Calendar Management System is grounded in the actual legislative workflow of the Valenzuela City Sangguniang Panlungsod, as previously outlined in the system flowchart (Figure 19) and the Business Process Architecture diagrams (Figures 12–13) presented in Chapter 3.7. The following core business processes were identified:

1. Calendar Management - Establishing and maintaining the master legislative calendar, including viewing existing schedules, checking for conflicts, and adding, editing, or synchronizing events.

2.Session Scheduling - Identifying available time slots, checking resource and participant availability, and generating session identifiers for plenary sessions, committee meetings, and public hearings.

3.Agenda Creation and Management - Defining agenda items, assigning owners and priorities, estimating durations, and establishing dependencies among items.

4.Deadline Tracking and Monitoring - Defining milestones, assigning responsibilities, and tracking task progress against legal deadlines, categorizing items as on schedule, at risk, critical, or overdue.

5.Document Intake, Analysis, and Classification - Receiving uploaded legislative documents (bills, resolutions, committee reports, minutes) and automatically classifying and summarizing them using AI-classification.

6.Reminder and Notification - Generating and dispatching automated alerts based on deadline status and upcoming events.

7.Deliberation and Voting - Supporting the recording of deliberation outcomes and voting results during sessions.

8.Action-Item Follow-up - Tracking action items generated from sessions through to completion.

9.Document Finalization and Archiving - Recording, tracking, and archiving finalized legislative outputs for long-term reference.

10.Post-Session Reporting - Updating calendar statuses, generating reports, and recording feedback after session completion.

Each process has a clear trigger, an assigned owner among the system's actors (Staff, Admin, Superadmin), and a defined outcome that feeds into the subsequent process, forming the continuous legislative lifecycle described in Chapter 3.7.

5.2 Business Process Diagrams

The system's business processes are represented through the diagrams already established in Chapter 3: the Business Process Architecture Level 0 diagram (Figure 12), which presents the legislative lifecycle at a high level — pre-legislative activities, core legislative operations, and post-legislative tracking — and the Business Process Architecture Level 1 diagram (Figure 13), which decomposes this into the detailed process flow of committee profiling, agenda creation, session scheduling, deliberation, voting, and document finalization. The Level 1 Data Flow Diagram (Figure 8/15) and

the system Flowchart (Figure 19) further illustrate how data and control flow between these processes.

To complement these, Figure 20 (Business Process Swimlane Diagram) illustrates the handoffs between actors across the legislative lifecycle:

• Staff lane: Encodes committee agendas, uploads documents, submits meeting requests.

• System/AI-Classification Lane: Automatically processes and classifies uploaded documents, generates summaries, and triggers downstream notifications.

• Admin lane: Encodes agendas, manages schedules, resolve conflicts flagged by the system.

• Superadmin lane: Oversees system configuration, monitors logs, and manages escalations.

5.3 Alignment of Integrated System with Business Process

Table 18: Alignment of Business Process with System Modules

Business Process	Supporting Modules	Problem Addressed (Table 1)
Calendar Management	Calendar Management Service	Agenda Scheduling
Session Scheduling	Session Scheduling Service	Agenda Scheduling, Activity Coordination
Agenda Creation and Management	Agenda Management Service	Agenda Scheduling, Information Access
Deadline Tracking and Monitoring	Deadline Tracking Service	Deadline Management
Document Analysis/ AI Classification	Document Analysis/AI Classification Service	Document Processing
Reminder and Notification	Reminder/Notification Service	Notification Systems
Deliberation and Voting	Session Scheduling Service, Agenda Management Service	Activity Coordination
Action-Item Follow-up	Deadline Tracking Service, Notification Service	Deadline Management, Notification Systems
Document Finalization and Archiving	Document Analysis Service, centralized document repository	Document Processing
Information Access Post-Session Reporting	Calendar Management Service, reporting layer	Information Access

This alignment demonstrates that every core legislative business process identified in Section 5.1 has a corresponding, purpose-built

module in the system architecture, and that each module directly addresses one or more of the operational problems documented during the requirements phase — ensuring the system is not simply a digitization exercise, but a process-aligned solution.

5.4 Business Process Improvement

Table 19: As-Is vs. To-Be Process Comparison

Process	As-Is Manual	To-Be System Supported	Related Object
Agenda Scheduling	Manual, inconsistent scheduling via email/paper, prone to conflicts	Centralized scheduling platform with automated conflict detection	Objective 1, 6
Deadline Management	No systematic tracking; missed legal deadlines and publication conflicts	Automated deadline tracking with categorized status (on schedule, at risk, critical, overdue)	Objective 2
Legislative Reading Stages	Manually tracked across separate documents	Integrated First/Second/Third Reading tracking tied to the calendar and agenda	Objective 3
Notification	Fragmented, unreliable alerts to stakeholders	Automated, rulebased notifications and reminders	Objective 4
Document Processing	Manual review, categorization, and	AI-classification and automated summarization of	Objective 5

	summarization by staff	legislative documents	
Conflict Prevention	Reactive; conflicts discovered after the fact	Proactive automated detection of scheduling and deadline conflicts	Objective 6

APPLICATON ARCHITECTURE

6.1 Components of Application Architecture

The Application Architecture of the system consists of the following components, consistent with the Microservice Architecture already presented in Figure 6 (Chapter 3.5):

• Presentation Layer — the web-based (and, where applicable, mobile-responsive) client interfaces used by Staff, Admin, Superadmin, and public Users.

• API Gateway — the centralized entry point for all client requests, responsible for authentication, SSL security, request routing, protocol translation, request aggregation, and audit logging (Figure 14).

• Core Microservices — Calendar Management, Session

Scheduling, Agenda Creation and Management, Deadline Tracking and Monitoring, Document Analysis/AI-Classification, and

Reminder/Notification, each operating as an independently deployable service (Chapter 3.5).

• Shared/Cross-Cutting Services — an Authentication and

Access Control service (supporting role-based permissions for Admin, Staff, Superadmin, and User), and an Audit Logging service that records transactions across all modules for accountability.

• Data Layer — the set of dedicated databases (Calendar DB, Session DB, Agenda DB, Deadline DB, Notification DB) described further in Chapter 7.

6.2 Application Architecture Diagram

The overall application structure builds on the Microservice Architecture diagram (Figure 6) and the Communication Pattern diagram (Figure 7) already presented in Chapter 3.5. Figure 21 below summarizes these into a single-layered application architecture view, showing the flow from client to gateway to service layer to data layer.

Unlike Figure 6, which was introduced in Chapter 3 primarily to justify the choice of a microservices approach for scalability and DevOps purposes, Figure 21 is intended to communicate the structural organization of the application for readers focused on how the system is built and how its parts relate to one another functionally.

6.3 Application Architecture Diagram

Software modules are integrated through well-defined interfaces exposed via the API Gateway, with each module retaining ownership of its own business logic and data, consistent with the database-per-service pattern implied by the module list in Chapter 3.5.

• The Agenda Management module exposes agenda identifiers and status to the Deadline Tracking module, which uses this data to establish and monitor deadline rules.

• The Document Analysis/AI-Classification module publishes classification results (document type, confidence score) and generated summaries, which are consumed by the Agenda Management module and attached to the relevant agenda item.

• The Session Scheduling module and Calendar Management module exchange session data bidirectionally to keep the master calendar and session records synchronized.

• The Reminder/Notification module subscribes to events from the Deadline Tracking and Session Scheduling modules to determine when alerts should be generated.

Shared data contracts between modules are kept intentionally minimal — each integration point passes only the identifiers and status fields required by the receiving module (e.g., agenda ID, deadline status, classification label) rather than full internal records, preserving each module's independence as described in Chapter 3.5.

This section covers integration within LACS 's own microservices. LACS also sits within a larger seven-system LGU 2 ecosystem (alongside ORTS, CMS, PHMS, PCMS, LRMS, and LAS), for which the integration contracts are external rather than internal and are treated separately in Chapter 4.4 and Appendix A.2.

6.4 Application Architecture Diagram

Building on the Communication Pattern diagram already introduced in Chapter 3.5 (Figure 7), the system uses a combination of synchronous (request/response) and asynchronous (event-driven) communication, selected according to the nature of each interaction:

• Synchronous communication is used where an immediate response is required for correct operation — for example, when the Session Scheduling service requests agenda data from the Agenda Management service, or when the Deadline Tracking service updates the Calendar Management service upon a deadline change (as illustrated in Figure 7).

• Asynchronous, event-driven communication is used for operations that do not require an immediate response and can tolerate background processing — most notably, the Document Analysis/AI-Classification module, which processes uploaded documents as a background job and publishes a "classification complete" event once processing finishes, rather than blocking the user's upload request.

• The Reminder/Notification service operates on a publish-subscribe pattern, listening for events from the Deadline Tracking and Session Scheduling services and generating notifications independently of the originating service's own processing.

This hybrid pattern ensures that time-sensitive operations (such as conflict detection during scheduling) remain responsive, while resource-intensive operations (such as AI-based document classification) do not degrade the performance of the rest of the system.

DATA ARCHITECTURE

7.1 Data Sources and Types

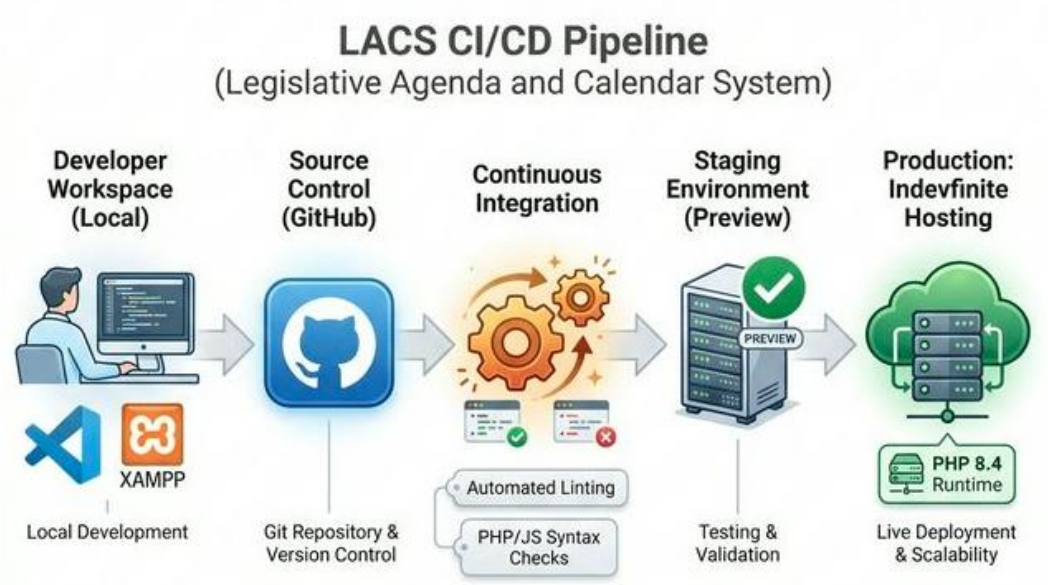
Table 20: Data Sources and Types

Data Category	Example	Structure	Sources
User-entered operational data	Agenda items, committee information, session details	Structured	Staff, Admin
Uploaded legislative documents	Bills, resolutions, committee reports, meeting minutes	Unstructured (text)	Staff
AI-generated data	Document classifications, generated summaries	Semi-structured	Document Analysis/AI Classification module
System generated data	Audit logs, notification logs, session timestamps	Structured	System (automated)
Reference/ master data	Calendar templates, deadline rules, user roles and permissions	Structured	Admin/ Superadmin configuration
Conflict Prevention	Reactive; conflicts discovered after the fact	Proactive automated detection of scheduling and deadline conflicts	Objective 6

7.2 Data Storage and Management

The system's core data flows are represented by the Level 0 (context) and Level 1 Data Flow Diagrams already presented in Chapter 3.7 (Figures 8 and 15). The Level 0 diagram defines the system boundary, showing Users, Staff, Admin, and Superadmin as external entities interacting with the system as a whole. The Level 1 diagram decomposes the system into its major processes — Calendar Management, Session Scheduling, Agenda Creation and Management, and Deadline Tracking and Monitoring — and their associated data stores (Calendar DB, Session DB, Agenda DB, Deadline DB, Notification DB).

To complete the data architecture picture, Figure 22 extends this with the data flow specific to the AI-classification pipeline, which was not detailed in the earlier diagrams:



This pipeline begins when a Staff member uploads a document, which is preprocessed (text extraction, cleaning) before being passed to the classification model. The model assigns a document type label and a confidence score, while a separate summarization step generates a condensed version of the document's content. Both outputs are then written back to the Agenda Database and linked to the corresponding agenda item, consistent with FR -INT-04 in Chapter 4.4.

7.3 Data Synchronization Across Systems

Because each microservice maintains its own database, the system relies on event-driven synchronization to keep related data consistent across services, following the asynchronous communication pattern described in Chapter 6.4. When a change occurs in one service — for example, a rescheduled session in the Session Scheduling service — an event is published and consumed by dependent services (Calendar Management, Deadline Tracking) to update their own records accordingly. This results in eventual consistency across the system: updates propagate within a short, acceptable window rather than instantaneously, which is an appropriate trade-off given the microservices architecture and is consistent with how Figure 7 (Chapter 3.5) already describes cross-service updates.

Synchronization also extends outward to the three LGU 2 systems directly connected to LAC S , as detailed in Chapter 4.4: OR TS , which pushes

ordinance proposal and reading-stage data into LACS and receives session-lock confirmations back; CMS, which pushes committee-routing deadlines and emergency meeting requests into LACS and receives deadline reminders and committee-membership updates back; and PHMS which receives confirmed hearing dates from LACS to generate public notices. Because these are separately developed systems within the same LGU 2 suite rather than internal LACS microservices, synchronization with them is treated the same as any external integration — mediated through the API Gateway (FR INT-14) rather than through the internal event bus used for LACS 's own modules (Chapter 6.4). LAC S does not synchronize directly with PCMS, LRMS, or LAS, since those systems consume data that already passed through ORTS, CMS, or PHMS rather than through LACS itself. Synchronization also covers the Calendar Export Functionality (Chapter 3.3 Scrum Board), which keeps legislative calendar data current in external calendar applications used by committee members (via iCalendar export, per FR -INT-13).

TECHNOLOGY ARCHITECTURE

8.1 Technology Stack and Infrastructure

The Technology Architecture defines the concrete technical foundation supporting the four architectural domains introduced in Chapter 3.8. The system was implemented using the following stack:

• Front-end: PHP, HTML, and JS, chosen for its component-based structure suited to a modular, multi-role interface (Staff, Admin, Superadmin, public User views).

• Back-end: [e.g., Node.js with Express / Django / Laravel — specify your actual framework], implementing each core microservice described in Chapter 6.

• Database engine(s): MYSQL is selected per service based on data structure needs (structured relational data for Calendar/Agenda/Deadline; potentially document-oriented storage for classification/summary outputs).

• AI-Classification/NLP component: [specify the model or library used — e.g., a fine-tuned transformer-based classifier, or a library such as spaCy/scikit-learn/Hugging Face Transformers], used to classify uploaded legislative documents and generate summaries as described in Chapter 7.2.

• Hosting/Cloud Infrastructure: [e.g., AWS / Google C loud / Azure / on-premises server — specify], supporting the C I/C D pipeline already described in Chapter 3.6 (Figure 9).

• Containerization: [e.g., Docker], used to package each microservice independently, consistent with the microservices architecture in Chapter 3.5.

• API Gateway implementation: [specify tool/framework, if any dedicated gateway product was used, e.g., Kong, NGINX, or a custom-built gateway], implementing the authentication, routing, and audit logging behavior described in Figure 14.

This stack was selected to directly support the scalability, independent deployment, and modularity goals established in the Microservices Architecture (Chapter 3.5) and to enable the automated CI/CD pipeline already documented in Chapter 3.6.

8.2 Software Technologies

Table 21: Software Technologies Used by Function

Function	Tool/Technology
Integrated Development Environment (IDE)	Visual Studio Code
Version Control	Github Repository
Project/Sprint Management	Monday.com
API Testing	AuthAPI
Diagramming	Lucidchart / draw.io
UI/UX Prototyping	Github Actions

Monitoring/Alerting	Monday.com

8.3 Scalability and Performance Consideration

The system's scalability rests on the microservices approach already justified in Chapter 3.5: because each service (Calendar Management, Session Scheduling, Agenda Management, Deadline Tracking, Document Analysis, Reminder/Notification) operates independently, individual services can be scaled based on their specific load characteristics. For instance, the Document Analysis/AI-Classification service — which is computationally heavier than the other services due to NLP processing — can be scaled independently of lighter services such as Calendar Management, without over-provisioning resources for the entire system.

Expected usage is bounded by the scale of a single LGU legislative office: a moderate number of concurrent internal users (Staff, Admin, Committee/Legislative Members) and a larger, but read-mostly, volume of public users accessing the transparency features. This usage pattern favors read-optimized scaling for public-facing endpoints and write optimized handling for internal transactional endpoints (agenda creation, deadline updates).

Performance visibility is supported by the Monitoring and Alerting architecture already described in Chapter 3.6 (Figure 11), which collects system metrics and logs across all microservices in real time and triggers alerts when predefined thresholds — such as elevated response times or error rates — are exceeded, enabling proactive performance management rather than reactive troubleshooting.

DEVELOPMENT PROCESS

9.1 Agile Scrum Roles and Responsibilities

The development of the system followed the Agile Scrum methodology introduced in Chapter 3.1, with roles and responsibilities distributed as previously defined in Table 2 (Chapter 3.2). Mr. R onald R oldan J r. served as Research Adviser, providing academic and technical oversight; Mr. J ojo Faustino served as Product Owner, representing the interests of the Valenzuela City Sangguniang Panlungsod and defining system requirements; Mr. J ohn Marvin I. Agno served as Project Manager, overseeing sprint planning, daily stand-ups, and sprint reviews; Mr. Shane Z. Salido served as Lead Programmer, responsible for core architectural design, backend development, database structure, and API integration; Mr. J aymar S . Rabino supported debugging, testing, and documentation; and

Ms. Jamille V. Francisco and Mr. Reinhard D. Ganal served as Documentation Specialists, responsible for diagrams, tables, and manuscript compilation.

Through the course of development, this role structure enabled clear ownership of each Scrum ceremony and artifact: the Project Manager maintained sprint cadence and removed impediments, the Lead Programmer maintained technical direction and code quality, and the Documentation Specialists ensured that architectural decisions and diagrams (Figures 1–22) were captured consistently as the system evolved.

9.2 Sprint Planning and Backlog Managements

Sprint planning was conducted at the start of each sprint cycle, during which the Project Manager and Product Owner reviewed the Product Backlog (Tables 6–9, covering EIS Information Security, EIS Standards, EIS Integration, and Analytics) and selected items for the upcoming Sprint Backlog (Table 10) based on priority and team capacity.

Backlog items were prioritized using a High/Medium priority scheme, as reflected in the Product Backlog tables, with prioritization criteria based on: (1) direct alignment with the six project objectives defined in Chapter 1.4, (2) dependency relationships (e.g., core authentication and calendar features were prioritized ahead of dependent features such as deadline

tracking), and (3) risk — features tied to legal compliance, such as deadline tracking and audit logging, were consistently assigned High priority given their connection to the Problem Statement (Table 1).

Backlog refinement occurred continuously: as sprints were completed, feedback from the Product Owner representative and observations from sprint reviews were used to add, re-prioritize, or split backlog items, consistent with the iterative requirements-gathering approach described in Chapter 4.2.

9.3 Sprint Execution and Deliverables

Table 22: Sprint Execution Summary

Sprint	Key Deliverables	Status
Sprint 1	User Authentication System, Basic Calendar Display, Session Timeout & Login Logging, Session Creation & Management, Data Encryption Implementation	Done
Sprint 2		
Sprint 3		
Final Sprint		

9.4 Sprint Planning and Backlog Managements

Several challenges were encountered during development, consistent with the complexity of building an AI-enhanced legislative system within a microservices architecture:

Integrating AI-classification with a transactional workflow. Because document analysis and classification are computationally heavier and less predictable in duration than standard transactional operations, the team had to design the Document Analysis module as an asynchronous, event-driven service (Chapter 6.4) rather than a synchronous one, to avoid blocking agenda-related workflows while classification was in progress.

Coordinating requirements validation with a government office. As the Product Owner representative, Mr. Jojo Faustino, was not available for continuous, day-to-day validation, the team relied on periodic sprint reviews rather than daily engagement, which occasionally slowed the feedback loop on newly implemented features. This was mitigated by batching related features for review together at defined checkpoints.

Managing cross-service data consistency. With each microservice maintaining its own database (Chapter 7.3), ensuring that related data — such as a rescheduled session correctly updating both the Calendar and Deadline services — remained consistent required careful event-driven design (Chapter 6.4, Chapter 7.4) and additional testing effort during integration (Chapter 10.4).

IMPLEMENTATION

10.1 Technical Implementation Details

The implementation phase translated the architecture defined in Chapters 6 through 8 into a working system. The Calendar Management, Session Scheduling, and Agenda Management services were implemented first, as they form the foundational data layer on which the Deadline Tracking and Reminder services depend, consistent with the dependency-based prioritization described in Chapter 9.2.

The Document Analysis/AI-Classification service was implemented as an independent, asynchronous microservice. Uploaded documents are queued for processing, preprocessed to extract clean text, passed through the classification model to determine document type, and summarized before the results are written back to the Agenda Database (Chapter 7.2, Figure 22).

The API Gateway (Figure 14) was configured to serve as the single-entry point for all client requests, implementing authentication and SSL enforcement before routing requests to the appropriate backend

microservice, and logging each transaction for the audit trail described in Chapter 6.1.

10.2 Technical Implementation Details

This section focuses on developer-facing tooling used to build the system, complementing the architecture-level technology choices already described in Chapter 8:

• Source control and repository management: [Git repository details — see Appendix A.13 for the full link and branching strategy]

• Package/dependency management: [e.g., npm/pip/composer — specify per your stack]

• Testing frameworks: [e.g., J est, PyTest, or your actual testing tools — used further in C hapter 11]

• Build and deployment scripts: [reference the C I/C D pipeline configuration detailed in Appendix A.14]

10.3 Code Integration and Interoperability

Independently developed microservices were integrated through the API contracts established during the design phase (Chapter 6.3). Each service exposes a defined interface — for example, the Document Analysis service exposes a standardized classification/summary output format

(document ID, classification label, confidence score, summary text) that the Agenda Management service consumes without needing knowledge of the underlying classification model.

Authentication across services is handled centrally through the API Gateway, so that individual microservices do not each implement their own authentication logic, reducing duplication and ensuring consistent access control across the system, consistent with the Shared/Cross-Cutting Services described in Chapter 6.1.

Data format standardization (e.g., consistent JS ON schemas for interservice events) was applied to reduce integration friction between services built at different points in the sprint schedule (Chapter 9.3).

10.4 Integration Testing and Debugging

Once individual modules were implemented, integration testing was performed to verify that modules functioned correctly together, distinct from the isolated unit testing covered in Chapter 11. Particular attention was given to:

• Agenda–Deadline integration: verifying that a newly created agenda item with an associated deadline correctly triggered the creation of a corresponding deadline record (FR -INT-02, Chapter 4.4).

• Document Analysis output integration: verifying that classification and summary outputs were correctly attached to their source agenda item,

including handling of edge cases such as documents that could not be confidently classified.

• Cross-service synchronization: verifying that changes such as a rescheduled session were correctly propagated to dependent services (Calendar, Deadline) within the expected synchronization window (Chapter 7.4).

Issues identified during integration testing — such as [describe your team's actual integration bugs, e.g., timing issues between asynchronous classification completion and agenda display, or mismatched data formats between services] — were logged, prioritized, and resolved prior to formal system testing (Chapter 11), with recurring issues informing the Challenges discussed in Chapter 9.4.

TESTING AND QUALITY ASSURRANCE

11.1 Testing Strategies and Methodologies

Testing was conducted across multiple levels to ensure the reliability of the system prior to deployment:

• Unit Testing — verifying individual functions and components within each microservice in isolation.

• Integration Testing — verifying that modules function correctly together, as described in Chapter 10.4.

• System Testing — verifying the system as a whole against the functional requirements defined in Chapter 4.

• User Acceptance Testing (UAT) — conducted with actual stakeholders from the Sangguniang Panlungsod, particularly Mr. Jojo Faustino and other office staff, to validate that the system met real operational needs identified in Chapter 4.1.

• AI-Classification Accuracy Validation — the Document Analysis/AI-Classification module was evaluated against a labeled set of sample legislative documents, measuring classification accuracy (and, where applicable, precision and recall) to ensure the model's outputs were reliable enough for operational use.

11.2 Test Cases and Test Data

Table 23: Sample Test Cases

Test Case ID	Module	Input	Expected Output	Actual Result	Pass/Fail
TC -01	Authentication	Valid Staff credentials	User is logged in and redirected to dashboard		
TC -02	Agenda Creation	New agenda item with title, owner, priority	Agenda item is created and stored		

TC -03	Deadline Conflict Detection	Two overlapping session requests	System flags scheduling conflict		
TC -04	Document Classification	Sample ordinance document upload	Document is classified as "Ordinance" with summary generated		
TC -05	Notification Dispatch	Deadline approaching threshold reached	Reminder notification sent to relevant user		
TC -06	Access Control	Non-admin user attempts admin action	Action is denied; access violation logged		

11.3 Test results and Bug Reports

Table 24: Notable Bugs and Resolutions

Bug ID	Module	Description	Severity	Resolution	Status
BUG-01					
BUG-02					

11.4 Quality Assurance Measures

Quality assurance practices were applied throughout development, not only at the testing phase:

• Code Review — all code changes were reviewed against the coding standards maintained by the Lead Programmer, as defined in the Development/Lead Programmer's responsibilities in Table 2.

• Security Testing — validated the Data Encryption Implementation, Comprehensive Audit Logging, and Granular Access Permissions features already identified in the Scrum Board and Product Backlog (Chapter 3.3, Table 6), confirming that role-based access restrictions correctly prevented unauthorized actions.

• Sprint-Level QA Sign-off — features were only marked "Done" on the Scrum Board (Chapter 3.3) after passing functional verification, ensuring that quality checks were embedded within the Scrum workflow rather than deferred to the end of the project.

• Documentation QA — the Documentation Specialists (Table 2) reviewed diagrams, tables, and written deliverables for accuracy and formatting consistency prior to inclusion in the manuscript.

RESULTS AND EVALUATION

12.1 Project Outcomes and Deliverables

12.2 Alignment with Project Objectives

Table 25: Objectives-to-Deliverables Traceability

Function	Tool/Technology
1. Centralized scheduling platform	Calendar Management and Session Scheduling modules
2. Automated deadline tracking and management	Deadline Tracking and Monitoring module
3. Integration of legislative reading stages with calendar/ agenda	Agenda Creation and Management module, integrated reading-stage tracking
4. Automated notification and reminder system	Reminder/Notification module
5. Automated document analysis and summarization using AIclassification	Document Analysis/AI Classification module
6. Conflict prevention mechanisms	Automated conflict detection within S ession Scheduling module

12.3 Stakeholder and User Feedback

12.4 Lesson Learned

At the project level, several lessons emerged that extend beyond the specific development challenges already discussed in Chapter 9.4:

On requirements gathering: relying primarily on a single key informant (Mr. Jojo Faustino) for requirements validation was efficient but occasionally limited the breadth of perspective available; involving a small additional group of Staff or Committee Members earlier in the process may have surfaced edge-case requirements sooner.

On architecture choices: the decision to adopt a microservices architecture with an asynchronous Document Analysis service (Chapter 6.4) proved well-suited to isolating the unpredictable processing time of AI-classification from the rest of the system, and is a pattern the team would recommend for similar AI-enhanced government systems.

On team coordination: the clear division of Scrum roles (Table 2) and the use of a visible Scrum Board (Chapter 3.3) helped maintain steady progress across sprints, particularly in surfacing blocked items early.

Chapter 5

CONCLUSION

13.1 Key Takeaways and Summary

The Valenzuela City Sangguniang Panlungsod's legislative operations were, prior to this project, constrained by manual and fragmented processes that led to scheduling conflicts, missed deadlines, unreliable notifications, and time-consuming manual document review, as established in the Problem S tatement (Chapter 1.3). This capstone project responded to that need by designing and developing the Local Government Unit 2: Legislative Agenda and Calendar Management System with Automated Document Analysis and Summarization Using AI-C lassification — an integrated, microservices-based system that centralizes legislative scheduling, automates deadline monitoring, and applies AI-classification to reduce the manual burden of legislative document review. Developed using the Agile Scrum methodology (Chapter 3), the system was designed, built, tested, and validated against the six objectives defined at the outset of the study, with results presented in Chapter 12 confirming that these objectives were met.

13.2 Key Takeaways and Summary

Beyond the immediate functional outcomes summarized in C hapter 12, this project offers two broader contributions consistent with the Significance and Relevance discussed in Chapter 1.5. First, it provides a practical, tested, and replicable model that other Philippine Local Government Units can reference when modernizing their own legislative operations, supported by the complete architectural documentation presented in Chapters 5 through 8. Second, it contributes a documented approach to combining a microservices architecture with AI-classification for a government workflow context — a combination that, as discussed in Chapter 2, remains an emerging area of study — offering Future Researchers and IT Professionals a concrete reference point for authentication, integration, and AI-adoption patterns in similar government information systems.

13.3 Future Works and Enhancements

Several directions for future work follow directly from the scope boundaries and lessons identified earlier in this manuscript:

• Deeper integration with the remaining LGU 2 ecosystem systems — PCMS, LRMS, and LAS — which LAC S does not

connect to directly today (Chapter 4.4). As the full seven-system LGU 2 suite (ORTS, LACS, CMS, PHMS, PCMS, LRMS, LAS) matures, extending LACS 's calendar and scheduling data further downstream (for example, surfacing session records directly within LRMS's archival views) would reduce reliance on ORTS, CMS,

and PHMS as intermediaries.

• Expansion into the areas explicitly excluded from this project's scope (Chapter 1.2) — financial management, performance analytics, and constituent engagement portals — represent natural next modules for a more comprehensive LGU information system.

• Mobile application development, to extend access beyond the current web-based interface for Staff and Committee Members who require on-the-go access to schedules and notifications.

• Continued refinement of the AI-classification model, particularly expanding the range of document types it can reliably classify and improving accuracy as more labeled legislative documents become available over time, building on the model evaluation approach described in Chapter 11.1.

13.4 Closing Remarks

This capstone project set out to address a real and clearly documented operational need within the Valenzuela City Sangguniang Panlungsod. Through the sustained application of the Agile Scrum methodology and a deliberately modular, integration-first architecture, the research team was able to deliver a system that not only meets its stated objectives but does so in a way that is extensible to the broader needs of Philippine local governance. It is the hope of the researchers that this work serves both the immediate needs of the Sangguniang Panlungsod and as a foundation for continued digital transformation within Philippine local government units.

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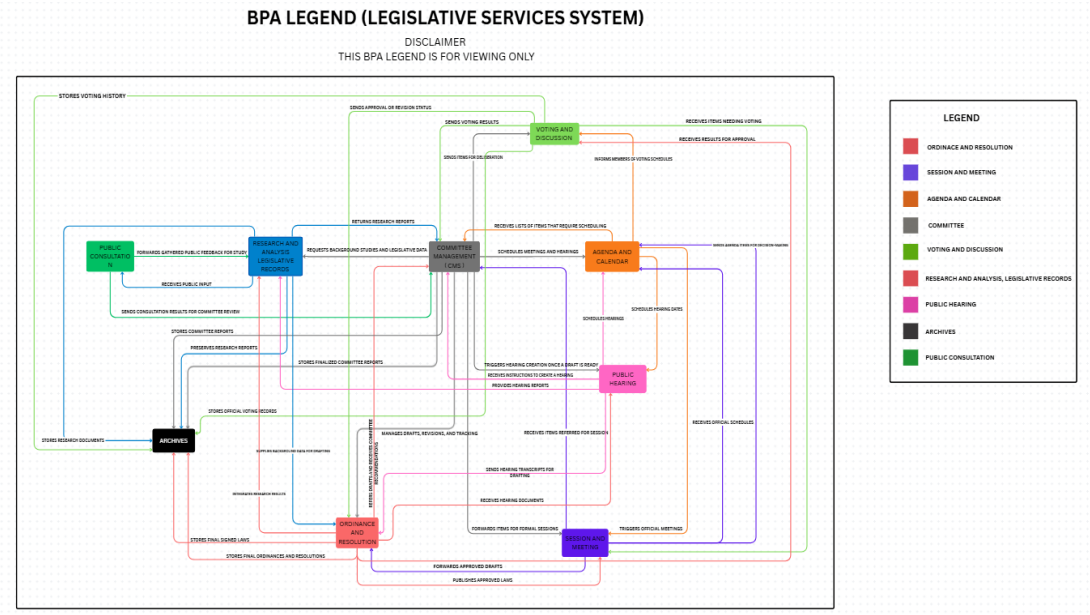
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APPENDICES

Appendix A: Detailed Technical Documentation

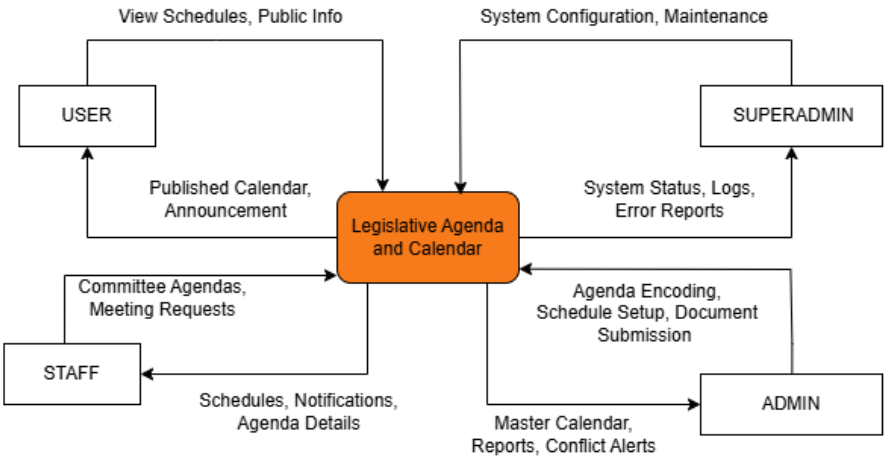
1. System Architecture

2. Information Systems Integration



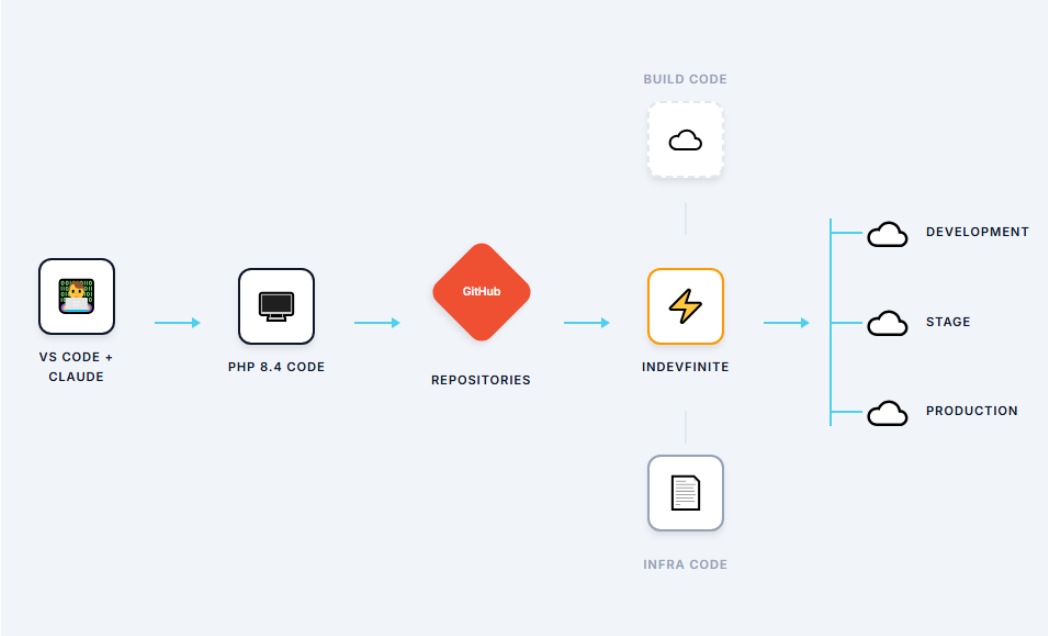
3. Application Design and Development

4. Database Schema and Data Management:



5. Network Configuration

6. Deployment and Infrastructure



7. Security Measures:

8. Testing and Quality Assurance

9. System Monitoring and Maintenance:

10. APIs and Integration Points:

11. User Documentation:

12. Known Issues and Troubleshooting

13. Version Control and Source Code Repository

14. DevOps and Continuous Integration/Continuous Deployment (CI/CD):

15. Licensing and Open Source Libraries:

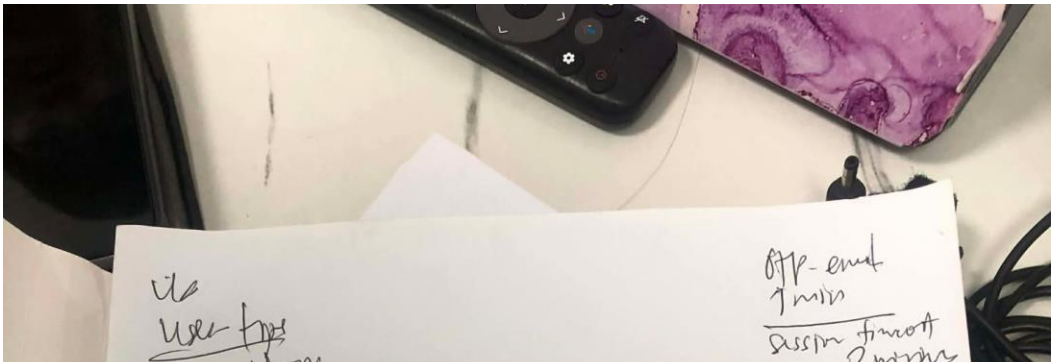
16. Performance Metrics and Monitoring:

Appendix B: Documents/URLs

1.Adviser Acceptance (signed)

2. Company Confirmation (signed)

3. Panel Evaluation and Signature (plus photo ops during defense)



4. Pilot Companies’ Background with proofs of interviews

5. Link to codes (good for 3 years)

6. Link to documentation (good for 3 years)

7. Link to demo video guide (good for 3 years

Appendix C: Certificate of Originality

Appendix D: Certificate of Grammarian

Name of Group Members/Individual:

1		4	
2		5	
3		6	

Title of the Project: EIS:

	Adviser	Grammarian
Name		
Designation		
Work Place Address		
Academic/ Professional Qualifications/ Membership		
Email Address		
Telephone		
Signature	*I hereby confirm that I have undertaken to supervise the project mentioned above and I do certify that I am not a member of the Project Examination Board (PEC)*	
	_____ *Date:*	_____ *Date:*



JOHN MERVIN I. AGNO

Project Manager

Contact

+639274321624

work.jmagno@gmail.com

Blk 79 Lot 84 Florante At Laura Street, Barangay Greater Lagro, Quezon City

Education

2023-2025
College
Bestlink Colleges of the Philippines

2021-2022
College
Technological Institute of the Philippines

2019- 2021
Senior High School
Lagro High School

2015-2019
High School
Lagro High School

Skills

- Mathematical Skills
- Creative idea
- Programming
- Logical Reasoning
- Leadership

Language

- English
- Filipino

About Me

I am a motivated and detail-oriented professional with experience in mathematics tutoring, web design, software development, and project management. I enjoy solving problems, building efficient digital solutions, and helping others learn and grow. With strong skills in PHP, JavaScript, and Python, I am passionate about creating user-friendly applications and continuously improving my technical and leadership abilities.

Experience

2024-2025
Project Manager (College Level)

Led academic projects by coordinating team members, defining objectives, managing timelines, and ensuring deliverables met quality standards. Facilitated communication, resolved challenges, and successfully completed projects within set deadlines.

2021 - 2023

Freelance Developer & Web Designer

Designed and developed custom websites and software solutions for clients across various industries. Gathered requirements, created wireframes and responsive designs, and implemented scalable, high-performance features. Demonstrated strong mastery of PHP, JavaScript, and Python to build dynamic web applications and backend systems. Managed projects from concept to deployment, delivering high-quality, client-focused digital solutions on time.

2017-2021

Mathematics Tutor/Adviser for Elementary School

Provided individualized and small-group instruction to elementary students, strengthening foundational math skills through engaging lessons, practice exercises, and progress tracking. Collaborated with parents and teachers to support academic improvement and build student confidence.



SHANE Z. SALIDO

Lead Programmer

Contact

+639623553420

salidoshane51@gmail.com

1620 Jasmin St. Area A Libis Brgy 175 Camarin Caloocan City

Education

2021- Present

College

Bestlink Colleges of the Philippines

2019- 2021

Senior High School

St. Clare College of the Philippines

2015-2019

High School

Camarin High School

Skills

Technical Skills

- Front-End Development: HTML, CSS, JavaScript
- Back-End Development: PHP
- Database Management: MySQL
- Programming Languages: Java, C++
- Development Tools: Visual Studio Code, Cursor
- Basic Network Cabling and Cable Management

Core Competencies

- Problem-solving and logical thinking
- System analysis
- Team Collaboration

About Me

I am a detail-oriented Fourth-year Bachelor of Science in Information Technology student with strong foundations in full-stack development and systems design. Proficient in Java, C++, HTML, JavaScript, PHP, and MySQL, I have developed academic projects applying structured development methodologies and research principles. Passionate about building efficient and user-centered systems, I am eager to contribute technical skills and continuous learning to real-world projects.

Certifications

October 19, 2025

Certificate of Participation

WordPress Web Development for Beginners & Online Businesses CloudTech Online Tutorial Services

July 13, 2025

Research Forum 2025

Theme: Empowering the future through Research: Local Solutions for Global Challenges

Bestlink College of the Philippines - Center for Research and development

August 3, 2025

Research Seminar 2025

Theme Starting the Research Journey: Building the Foundations for Academic and Social Impact

Bestlink College of the Philippines - Center for Research and Development



JAMILLE V. FRANCISCO

Document Specialist

Contact

+639677156593

jamillefrncsc@gmail.com

24 F.Carlos Baesa Quezon City

Education

2021 - Present
College
Bestlink College of the Philippines

2019-2021
Senior High School
San Francisco High School

2015-2019
High School
San Francisco High School

Skills

- Oriented in Microsoft Office Application such as:
  Microsoft Office Word
  Microsoft Excel
- Creativity
- Critical Thinking

Language

- Filipino
- English

About Me

Bachelor of Information Technology student with a focus on Information Security. I consider myself a responsible, organized, and committed to delivering quality work. I am looking forward to gaining my first hands-on experience in the IT field.

Certification

2025
Research Forum
"Empowering the future through Research Local Solutions for Global Challenges"

2025
Research Seminar
"Starting the Research Journey: Building the Foundations for Academic and Social Impact"



REINHARD D. GANAL

Document Specialist

Contact

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reinharddelacruzganal@gmail.com

116 Taurus St. Remarville Subd. Brgy. Bagbag Novaliches Quezon City

Education

2021 - Present
College
Bestlink College of the Philippines

2019-2021
Senior High School
Trinity Polytechnic College

2019
High School
San Bartolome High School

Skills

- Oriented in Microsoft Office Application such as:
  Microsoft Office Word
  Microsoft Excel
- Creativity
- Critical Thinking

Language

- Filipino
- English

About Me

Fourth-year Bachelor of Information Technology student with a focus on Information Security and technical documentation. Detail-oriented and committed to continuous learning, with a strong eagerness to apply and enhance technical skills in a professional environment.

Certification

2025
Research Forum
"Empowering the future through Research Local Solutions for Global Challenges"

2025
Research Seminar
"Starting the Research Journey: Building the Foundations for Academic and Social Impact"



JAYMAR S. RABINO

Project Developer

Contact

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Blk 35 Lot 5 Phase 4 Villa Teresita Alisasis, Barangay 171, Bagumbong, Caloocan City

Education

2023 - Present
College
Bestlink Colleges of the Philippines

2022-2023
College
St. Clare College

2019-2021
Senior High School
Vince Malapitan Senior High School

2016-2020
High School
Bagumbong High School

Skills

- Technical Skills
- Project Management
- Critical Thinking
- Leadership

Language

- Filipino
- English

About Me

Motivated and hardworking student with strong communication and problem-solving skills. Experienced in school projects, leadership activities, and basic computer applications. Eager to apply knowledge and learn new skills in a professional work environment.

Certification

2023
Maintenance in church
Working as maintenance in church in 3years and scholar of the priest.